Assessment Task 2

[Data exploration and preparation]

31250 Introduction to Data Analytics

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A. Initial Data Exploration

1. Attribute types

Attribute Name	Attribute Type	Justification
Date	Ordinal	Ordered by date but intervals are not equal. Specifically, Dates are ordered chronologically, making the sequence meaningful. However, the intervals between dates are not equal, especially when considering varying days between months and years. As demonstrated in Figure 1, this characteristic supports treating Date as an ordinal attribute, where the order is essential, but the intervals between dates are not consistent.
Location	Nominal	Names of places, Categorical data with no order
MinTemp	Interval	Measured in degrees Celsius with equal intervals between values, but no true zero
MaxTemp	Interval	Measured in degrees Celsius with equal intervals between values, but no true zero
Rainfall	Ratio	Measured in millimeters with a true zero, making it a ratio attribute.
Evaporation	Ratio	Measured in millimeters with a true zero, indicating no evaporation occurred
Sunshine	Ratio	Measured in hours with a true zero (no sunshine)
WindGustDir	Nominal	Direction is categorical with no order
WindGustSpeed	Ratio	Measured in kilometers per hour, with a true zero

WindDir9am	Nominal	Categorical directions without inherent order
WindDir3pm	Nominal	Categorical directions without inherent order
WindSpeed9am	Ratio	Measured in kilometers per hour with a true zero
WindSpeed3pm	Ratio	Measured in kilometers per hour with a true zero
Humidity9am	Ratio	Measured in percentage with a true zero
Humidity3pm	Ratio	Measured in percentage with a true zero
Pressure9am	Ratio	Measured in hectopascals, with a true zero
Pressure3pm	Ratio	Measured in hectopascals, with a true zero
Cloud9am	Ratio	Measured in eighths (fractions) with a true zero
Cloud3pm	Ratio	Measured in eighths (fractions) with a true zero
Temp9am	Interval	Measured on a scale with equal intervals without a true zero point
Temp3pm	Interval	Measured on a scale with equal intervals without a true zero point
RainToday	Nominal	Categorical data, specifically a binary categorical attribute (Yes/No) with no order
RainTomorrow	Nominal	Categorical data, specifically a binary categorical attribute (Yes/No) with no order

2. The summarising properties for the attributes

Frequency & Distribution Data Visualisations

2.1 Date (Ordinal)

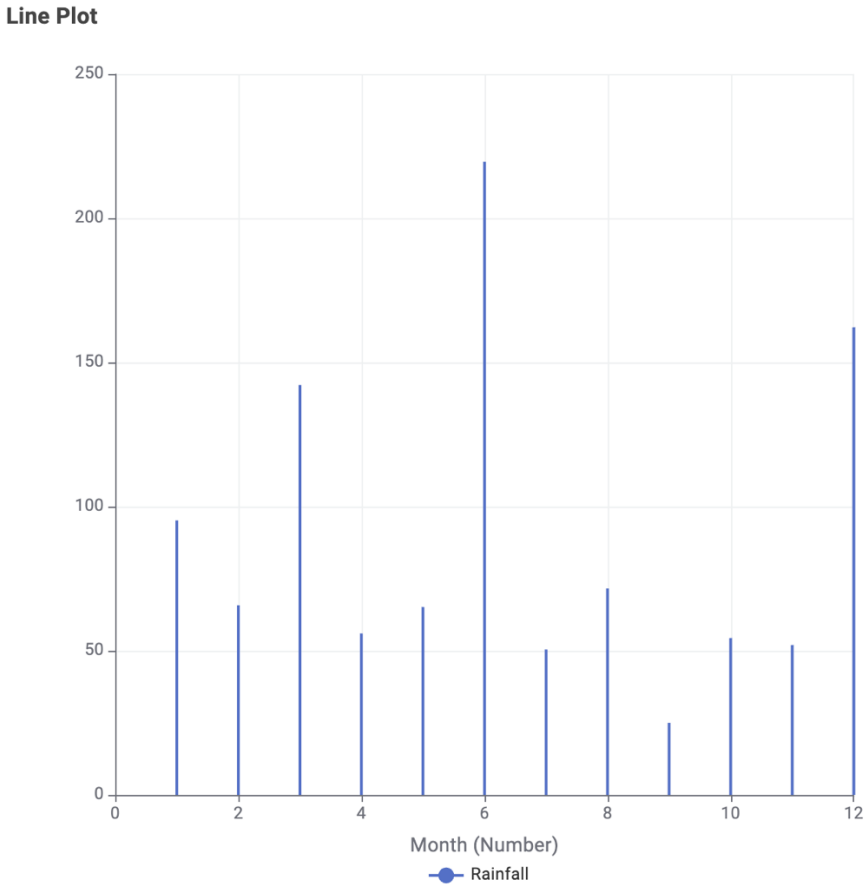


Figure 1: Line Plot for Rainfall across Month (Date)

The line plot above (refer to figure 1) represents the total rainfall distributed across the months of the year. The months are indicated numerically on the X-axis, with 1 representing January 2 representing February, and so on. The Y-axis displays the total amount of rainfall, measured in millimeters. A clear pattern of variation in rainfall is observed across the year, with the highest peak occurring in June (Month 6), where the rainfall exceeds 200 millimeters. This indicates that June is likely the peak of the rainy season in the region under study. Other months, such as February (Month 2) and December (Month 12), also show significant rainfall, although not as high as June, marking them as secondary peaks. Conversely, months like September (Month 9) and April (Month 4) experience the lowest levels of rainfall, suggesting these could be part of a dry season.

2.2 Location (Nominal)

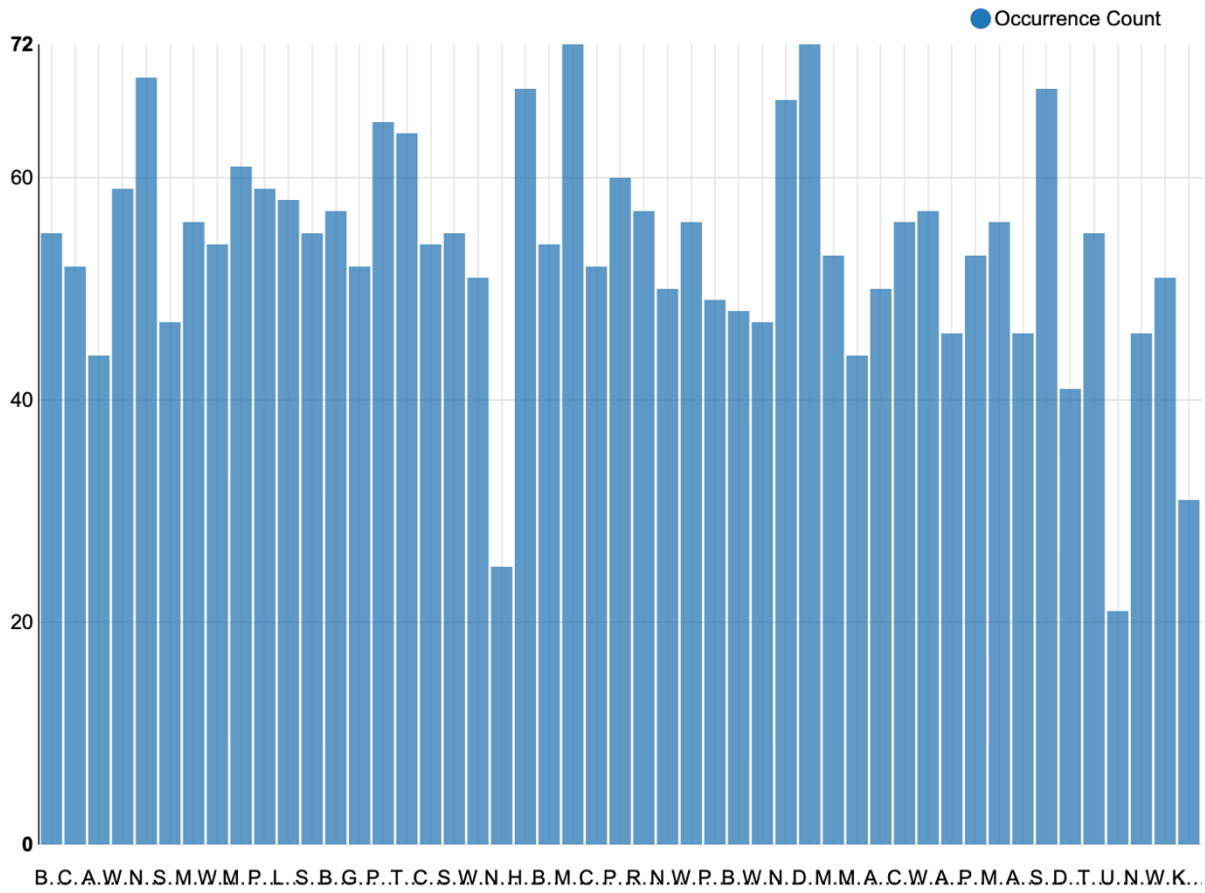


Figure 2: Bar Chart for Location

The bar chart above (refer to figure X) illustrates the occurrence count for various locations within the dataset. Each bar represents a unique location, and the height of the bar reflects the frequency with which that location appears in the dataset. From the chart, it's evident that locations like Dartmoor and Melbourne have the highest occurrence count, each registering a total of 72 occurrences. On the other hand, locations such as Nhil and Uluru exhibit much lower occurrence counts, with only 25 and 21 occurrences, respectively. This could suggest a geographical bias or specific focus areas within the data collection process, warranting further investigation into the reasons behind these variations

2.3 MinTemp (Interval)

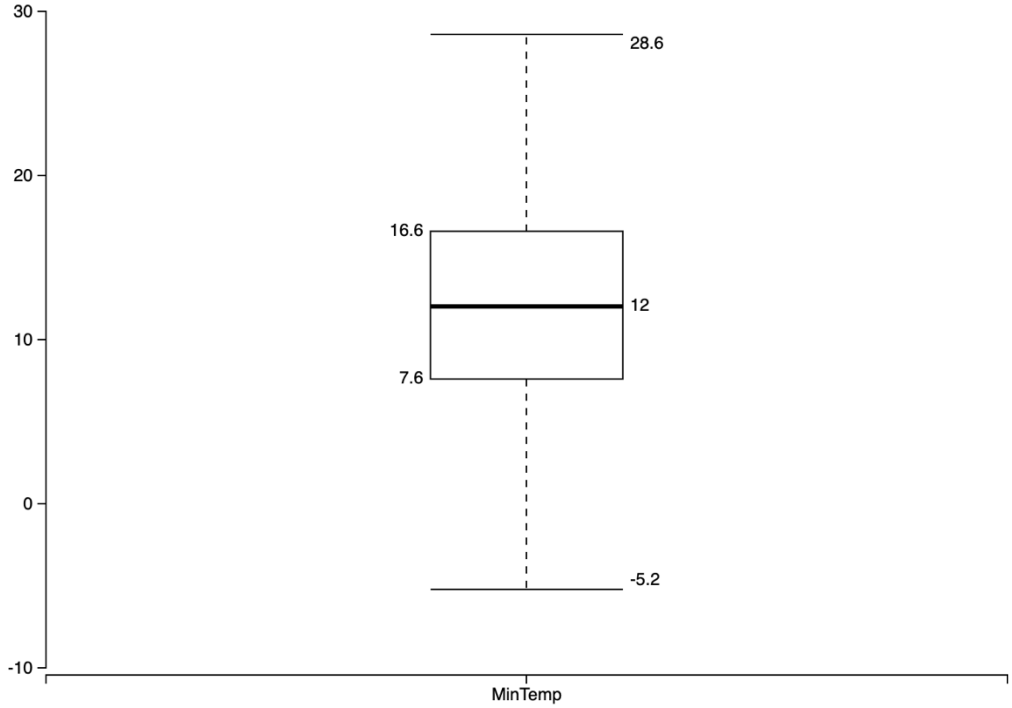


Figure 3: Box Plot Minimum Temperature (MinTemp)

The box plot above (refer to figure 2) represents the distribution of minimum temperatures (MinTemp) recorded in the dataset. The temperatures are indicated numerically on the Y-axis, with the values ranging from approximately -5.2°C to 28.6°C. The box in the plot spans from 7.6°C to 16.6°C, representing the interquartile range (IQR), where the middle 50% of the data lies. The central line within the box marks the median value at 12°C, indicating that half of the recorded minimum temperatures are below 12°C and half are above this value.

Statistics	Value
Mean	12.022°C
Median	12°C
Minimum Value	-5.2°C
Maximum Value	28.6°C
25% Quantile	7.6°C
75% Quantile	16.6°C
Standard Deviation	6.348
Variance	40.298

2.4 MaxTemp (Interval)

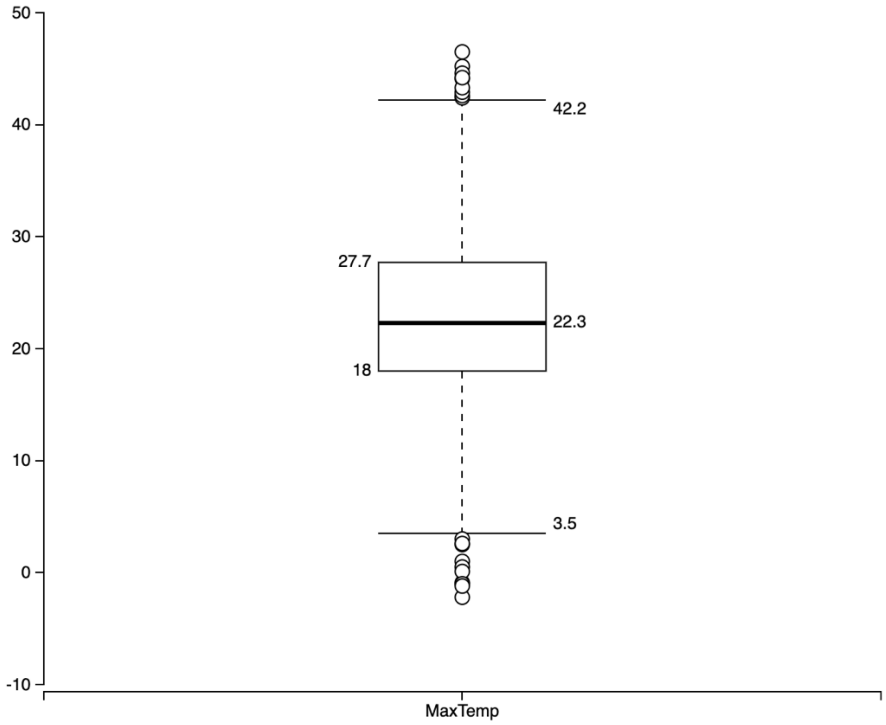


Figure 4: Box Plot Maximum Temperature (MaxTemp)

Statistics	Value
Mean	22.871°C
Median	22.3°C
Minimum Value	-2.2°C
Maximum Value	46.5°C
25% Quantile	18°C
75% Quantile	27.7°C
Standard Deviation	7.06
Variance	49.838

The box plot above (Figure 3) illustrates the distribution of maximum temperatures (*MaxTemp*) in the dataset, ranging from approximately 3.5°C to 42.2°C. The interquartile range (IQR) spans from 18°C to 27.7°C, with a median of 22.3°C, indicating that half of the data falls below this value and half above. Outliers above 42.2°C suggest rare extreme heat events, while those below 3.5°C indicate unusually cold days. This variability highlights the occurrence of both hot and cold extremes within the dataset.

2.5 Rainfall (Ratio)

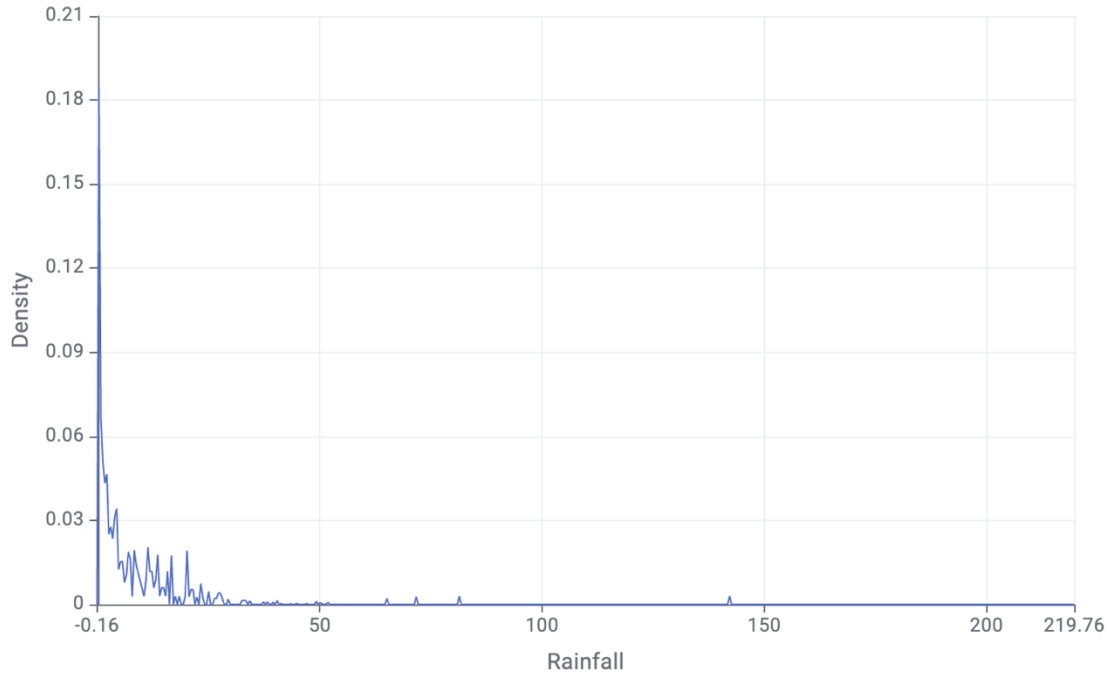


Figure 5: Density Plot for Rainfall

The density plot above (Figure X) shows the distribution of rainfall. The X-axis represents rainfall in millimeters, and the Y-axis shows density, indicating the frequency of rainfall values. The plot is highly skewed, with most rainfall amounts clustered near 0 mm, highlighting that low rainfall is common. The density drops sharply as rainfall increases, with a long tail showing occasional higher rainfall events. This indicates that while the area typically experiences low rainfall, there are rare instances of significant rainfall. The plot effectively visualizes the distribution and outliers in the rainfall data.

Statistics	Value
Mean	2.538
Median	0
Minimum Value	0
Maximum Value	219.6
25% Quantile	0
75% Quantile	0.8
Standard Deviation	9.574
Variance	91.652

2.6 Evaporation (Ratio)

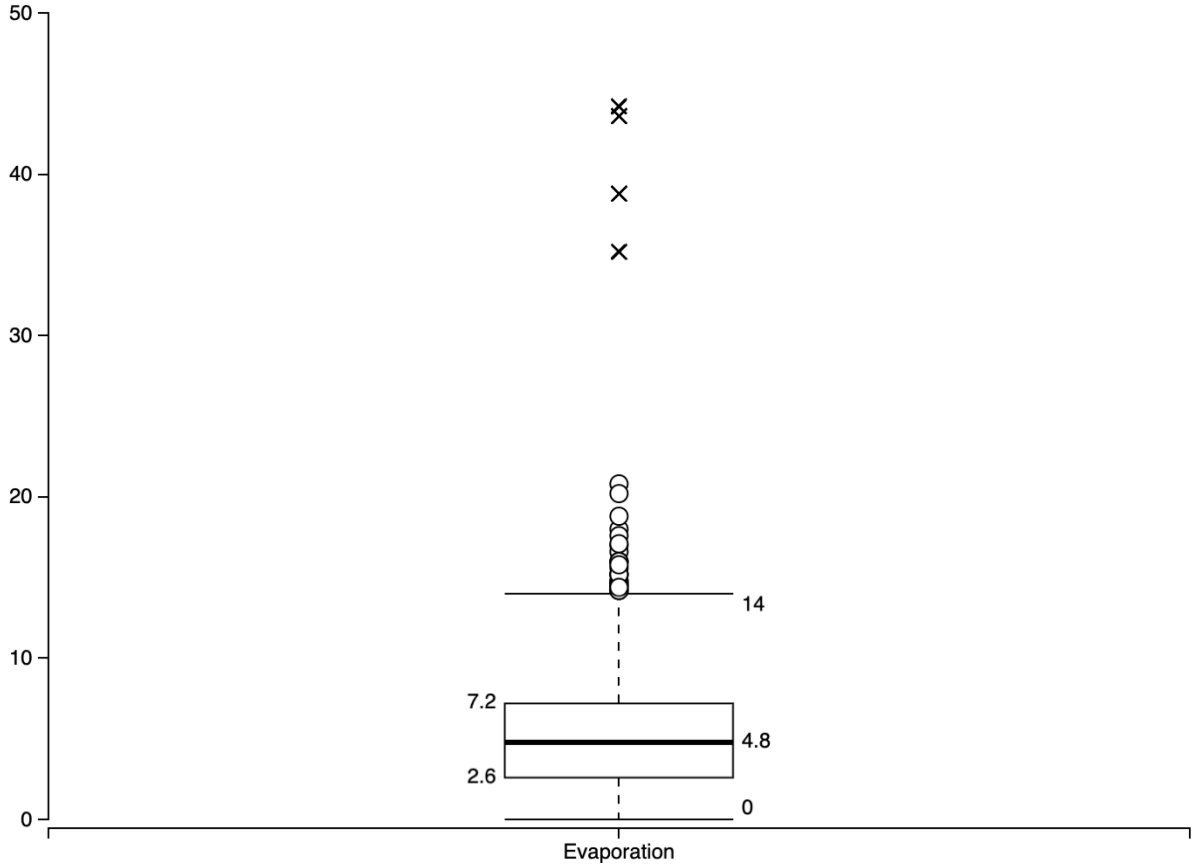


Figure 6: Box plot Evaporation

The above box plot of Evaporation reveals that most data points are concentrated within a lower range, with a median value of 4.8 mm. The interquartile range (IQR) spans from 2.6 mm to 7.2 mm, indicating that the middle 50% of the data falls within this range. The presence of several outliers above 14 mm suggests occasional spikes in evaporation, possibly due to specific weather conditions. These outliers, extending up to nearly 50 mm, highlight the variability in the data.

Statistics	Value
Mean	5.351
Median	0
Minimum Value	0
Maximum Value	44.2
25% Quantile	2.6
75% Quantile	7.2
Standard Deviation	3.857
Variance	14.873

2.7 Sunshine (Ratio)

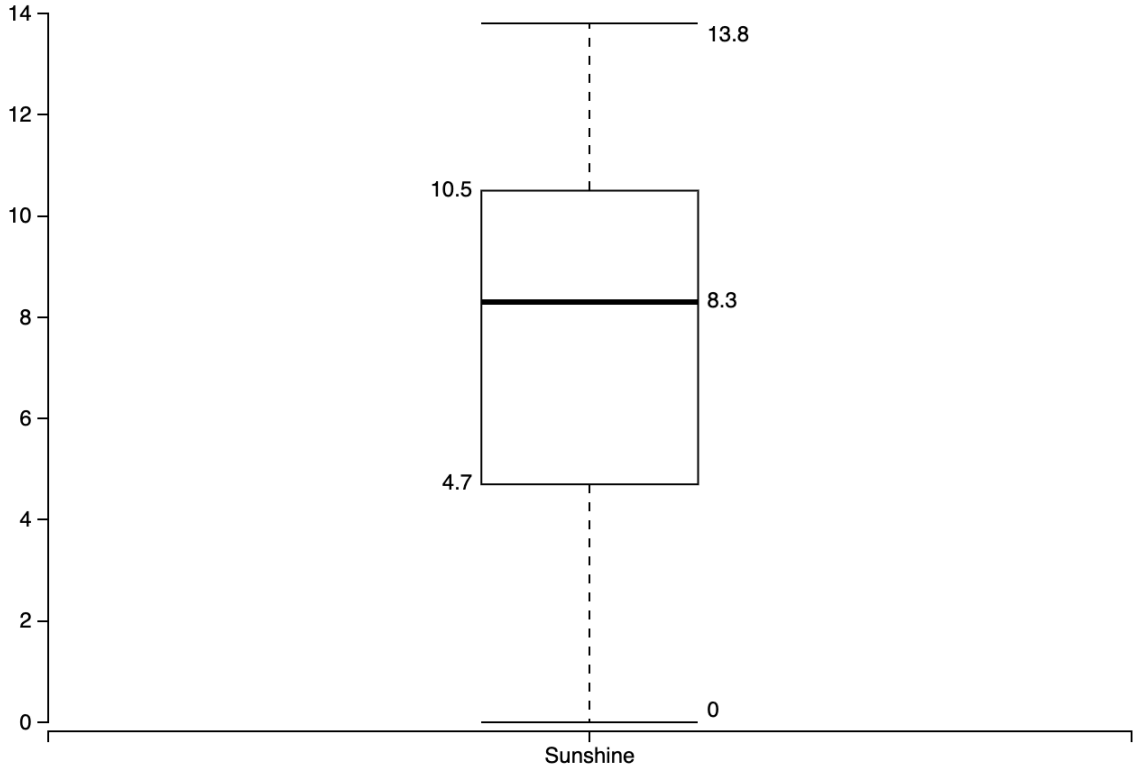


Figure 7: Box plot Sunshine

The above box plot of Sunshine indicates a median value of 8.3 hours. The interquartile range (IQR) extends from 4.7 hours to 10.5 hours, showing that the majority of sunshine data lies within this range. The maximum observed value reaches 13.8 hours, indicating some days with prolonged sunshine. Unlike the Evaporation plot, there are no extreme outliers, suggesting a more consistent pattern of sunshine across the dataset.

Statistics	Value
Mean	7.531
Median	0
Minimum Value	0
Maximum Value	13.8
25% Quantile	4.7
75% Quantile	10.5
Standard Deviation	3.831
Variance	14.677

2.8 WindGustDir (Nominal)

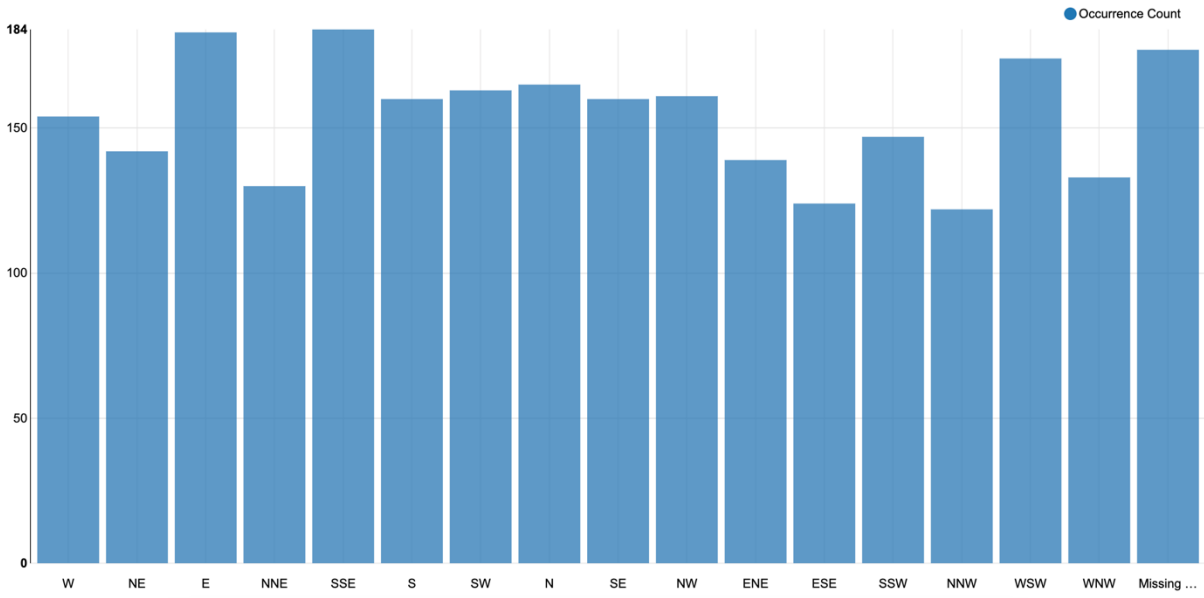


Figure 8: Bar chart for Wind Gust Direction (WindGusDir)

The bar chart above represents the occurrence count of various wind gust directions. The X-axis displays the wind directions, while the Y-axis indicates the number of occurrences for each direction. The wind gust directions "SSE" and "WNW" have the highest occurrence counts, suggesting these are the most frequently observed wind directions in the dataset. Conversely, directions like "NNE" and "SSW" show lower frequencies, indicating they are less common. The distribution across directions is fairly even, with no single direction significantly outweighing the others in occurrence. A "Missing" category is present, indicating that some data points lack a recorded wind gust direction, which may need to be considered in further analysis

2.9 WindGustSpeed (Ratio)

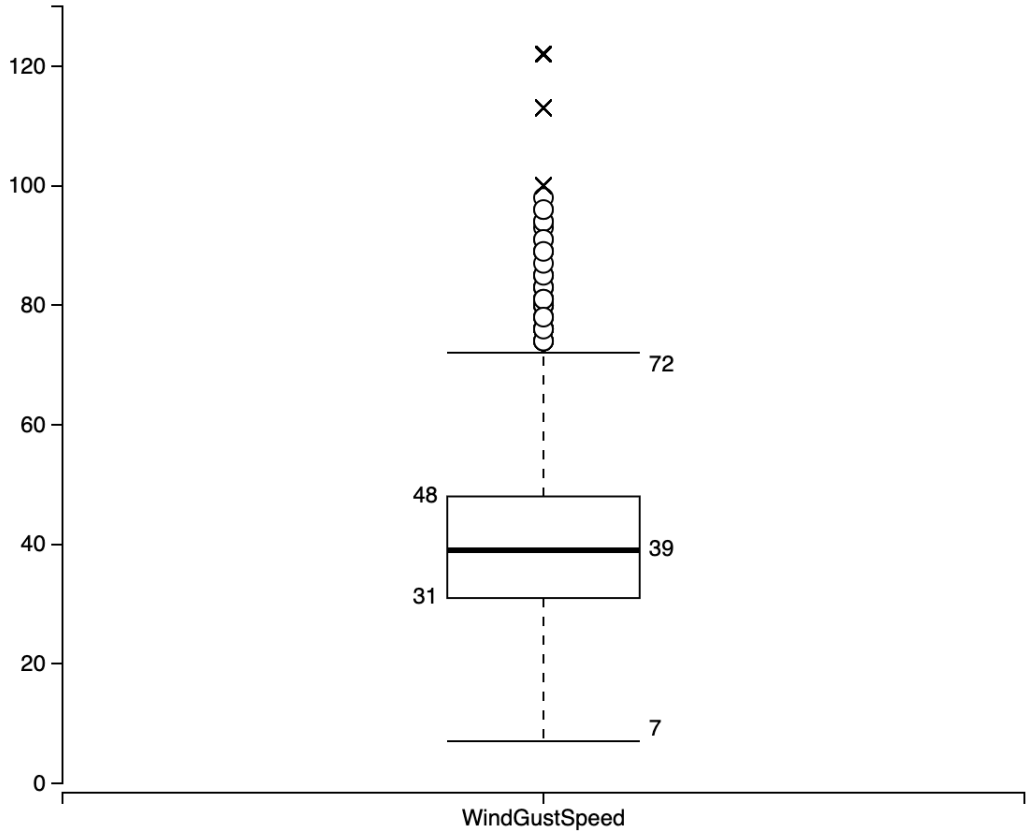


Figure 9: Box plot for Wind Gust Speed

The above box plot of Sunshine indicates a median value of 8.3 hours. The interquartile range (IQR) extends from 4.7 hours to 10.5 hours, showing that the majority of sunshine data lies within this range. The maximum observed value reaches 13.8 hours, indicating some days with prolonged sunshine. Unlike the Evaporation plot, there are no extreme outliers, suggesting a more consistent pattern of sunshine across the dataset.

Statistics	Value
Mean	40.18
Median	39
Minimum Value	7
Maximum Value	122
25% Quantile	31
75% Quantile	48
Standard Deviation	13.975
Variance	195.294

2.10 WindDir9am (Nominal)

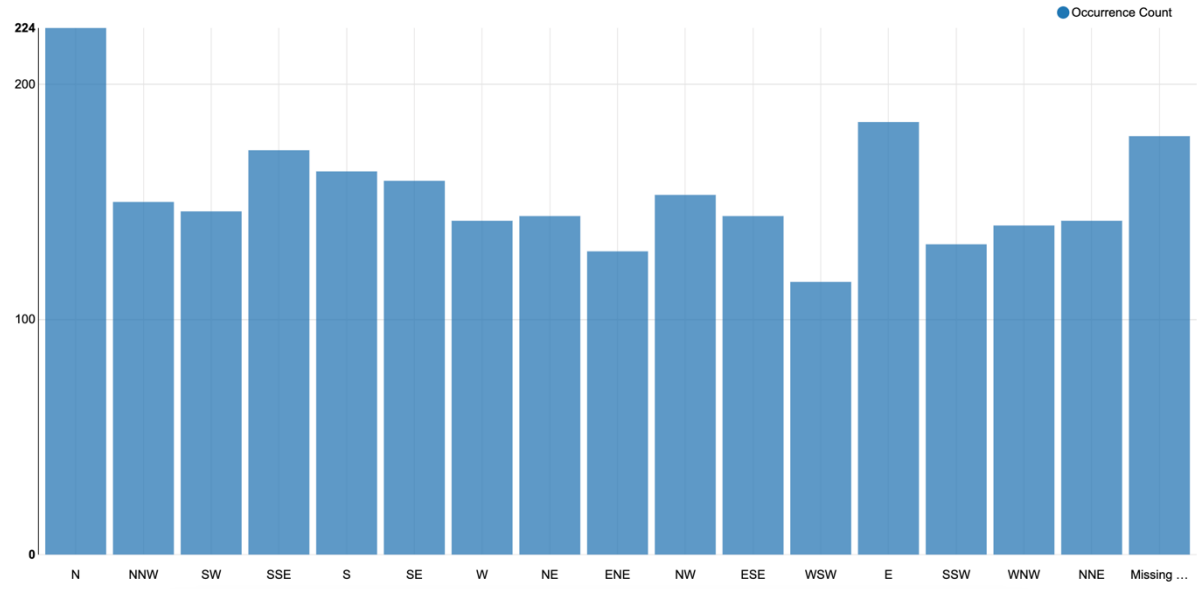


Figure 10: Bar chart for Wind Direction 9am (WindDir9am)

The bar chart for WindDir9am shows the distribution of wind directions recorded at 9 AM. The most common wind direction at this time is from the north (N), with an occurrence count of 224, making it the dominant direction in the morning. Other directions like NW and WSW also show significant occurrences, indicating they are common as well. On the other hand, directions such as SW and S have fewer occurrences, suggesting that these directions are less frequent in the morning. This pattern could reflect regional wind patterns influenced by various factors such as geography and weather conditions.

2.11 WindDir3Pm (Nominal)

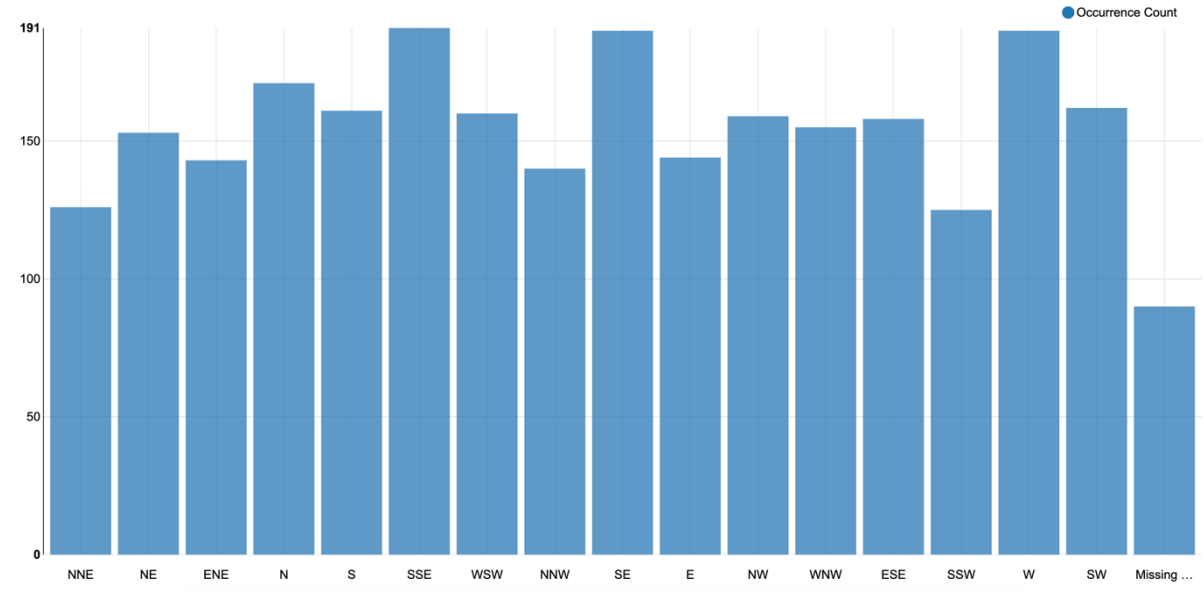


Figure 11: Bar chart for Wind Direction 3pm (WindDir3pm)

The bar chart for WindDir3pm reveals the distribution of wind directions recorded at 3 PM. The directions with the highest occurrence are WNW and NNW, each peaking around 190 occurrences, indicating that these directions are predominant during the afternoon. On the other hand, directions like NNE and SW have fewer occurrences, showing a less frequent wind direction at that time. This distribution suggests that certain wind directions are more common in the afternoon, potentially influenced by regional weather patterns or geographical factors.

2.12 WindSpeed9am (Ratio)

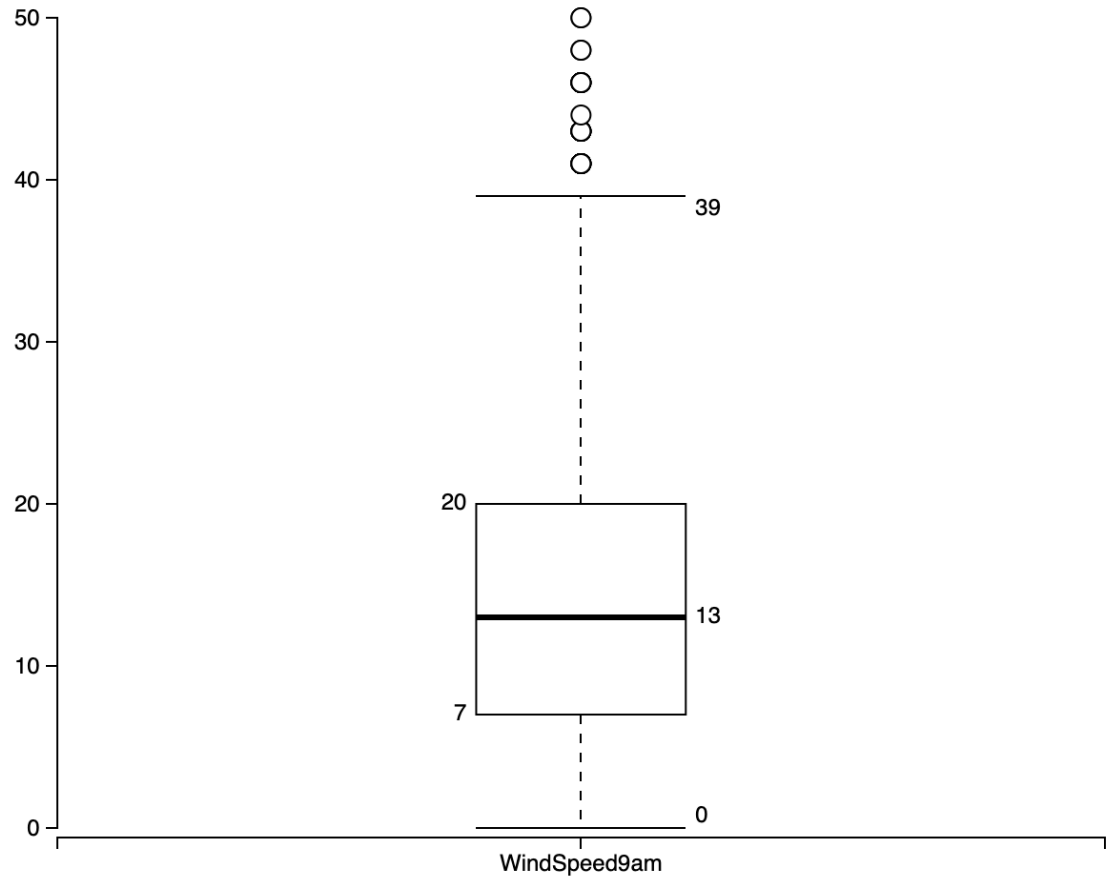


Figure 12: Box plot for Wind Speed 9am

The box plot for WindSpeed9am reveals that the median wind speed at 9 am is 13 units. The interquartile range (IQR) stretches from 7 to 20 units, indicating that the middle 50% of wind speeds fall within this range. The presence of several outliers above 39 units suggests that on some days, the wind speed at 9 am reaches significantly higher levels than usual, though these occurrences are relatively rare

Statistics	Value
Mean	14.145
Median	13
Minimum Value	0
Maximum Value	50
25% Quantile	7
75% Quantile	20
Standard Deviation	8.909
Variance	79.375

2.13 WindSpeed3pm (Ratio)

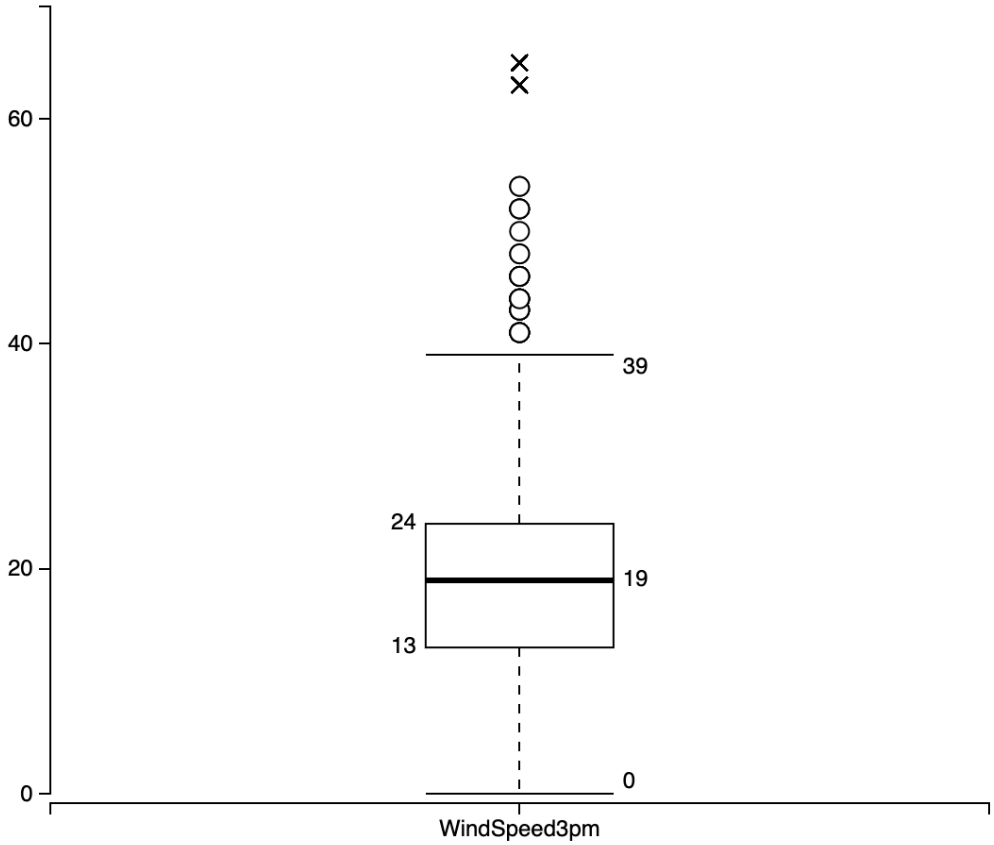


Figure 13: Box plot for WindSpeed3pm

The box plot for WindSpeed9am reveals that the median wind speed at 9 am is 13 units. The interquartile range (IQR) stretches from 7 to 20 units, indicating that the middle 50% of wind speeds fall within this range. The presence of several outliers above 39 units suggests that on some days, the wind speed at 9 am reaches significantly higher levels than usual, though these occurrences are relatively rare

Statistics	Value
Mean	18.657
Median	19
Minimum Value	0
Maximum Value	65
25% Quantile	13
75% Quantile	24
Standard Deviation	9.11
Variance	82.985

2.14 Humidity9am (Ratio)

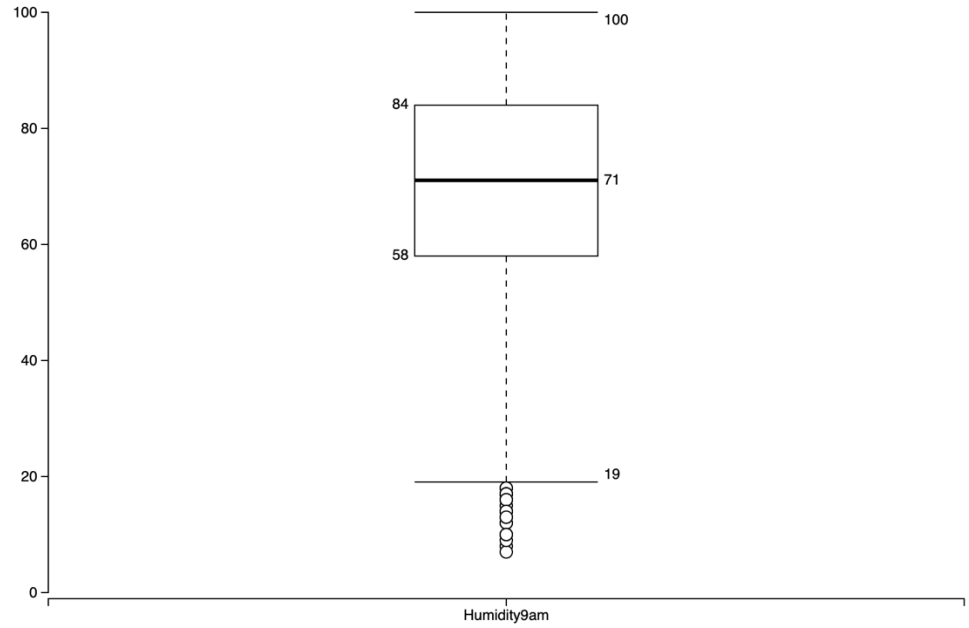


***Figure 14: Box plot of* Humidity9am**

In the box plot for Humidity at 9 AM, the median humidity is around 71%, suggesting that the mornings tend to be more humid. The interquartile range (IQR) extends from about 58% to 84%, showing a narrower concentration of data in the middle 50%. However, there are several outliers observed on the lower end, with humidity values as low as 19%, indicating some instances of unusually low morning humidity. The overall range extends from 19% to 100%.

Statistics	Value
Mean	69.504
Median	71
Minimum Value	7
Maximum Value	100
25% Quantile	58
75% Quantile	84
Standard Deviation	18.712
Variance	350.135

2.15 Humidity3pm (Ratio)

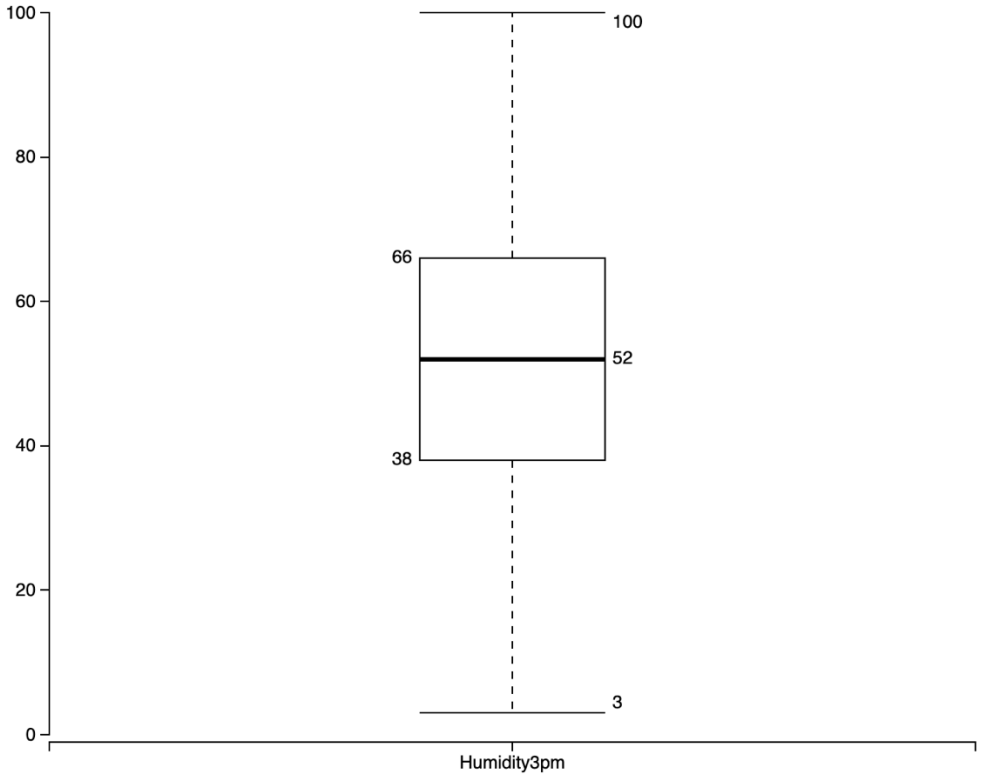


***Figure 15: Box plot of* Humidity3pm**

In the box plot for Humidity at 3 PM, the median humidity is approximately 52%, indicating that half of the observations have a humidity level below this value. The interquartile range (IQR) spans from about 38% to 66%, suggesting that the middle 50% of the data is concentrated within this range. Notably, the humidity values range from a minimum of 3% to a maximum of 100%, with no visible outliers.

Statistics	Value
Mean	51.983
Median	52
Minimum Value	3
Maximum Value	100
25% Quantile	38
75% Quantile	66
Standard Deviation	20.534
Variance	421.65

2.16 Pressure9am (Ratio)

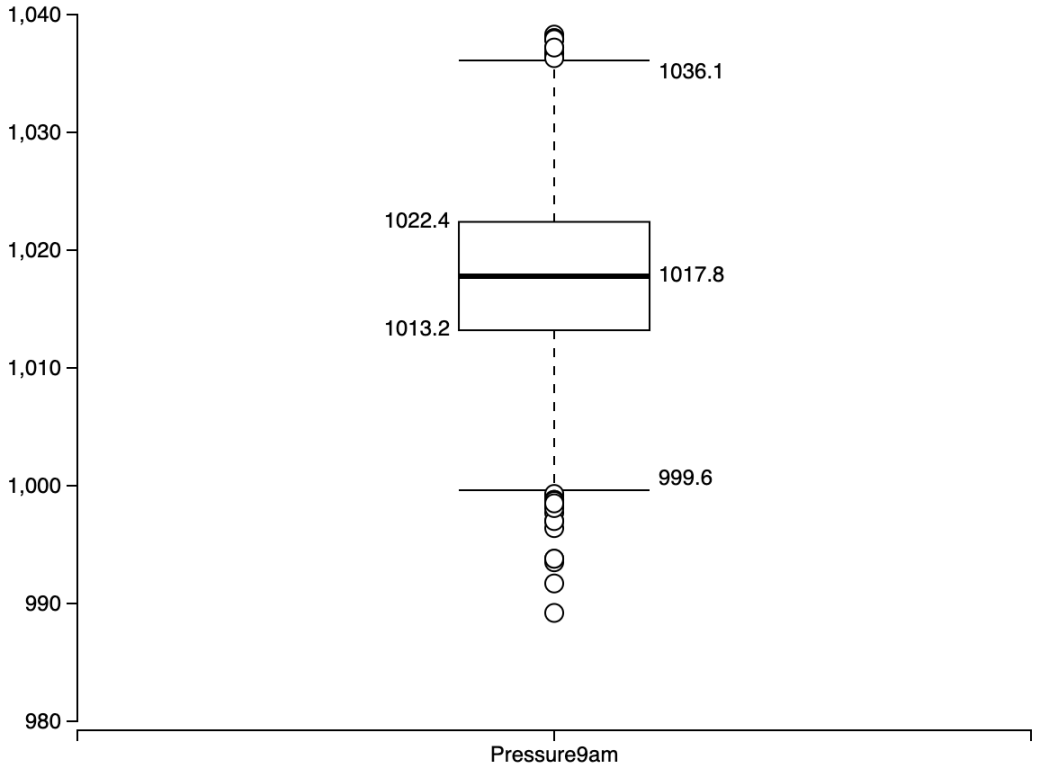


Figure 16: Box plot for Pressure9am

The above box plot of Pressure9am indicates a median value of 1017.8 hPa. The interquartile range (IQR) extends from 1013.2 hPa to 1022.4 hPa, showing that the majority of the pressure data at 9 am lies within this range. The maximum observed value reaches 1036.1 hPa, with several outliers below 999.6 hPa, suggesting some variability in atmospheric pressure during this time.

Statistics	Value
Mean	1017.828 hPa
Median	1017.8 hPa
Minimum Value	989.2 hPa
Maximum Value	1038.3 hPa
25% Quantile	1013.2 hPa
75% Quantile	1022.4 hPa
Standard Deviation	7.003 hPa
Variance	49.047 hPa

2.17 Pressure3pm (Ratio)

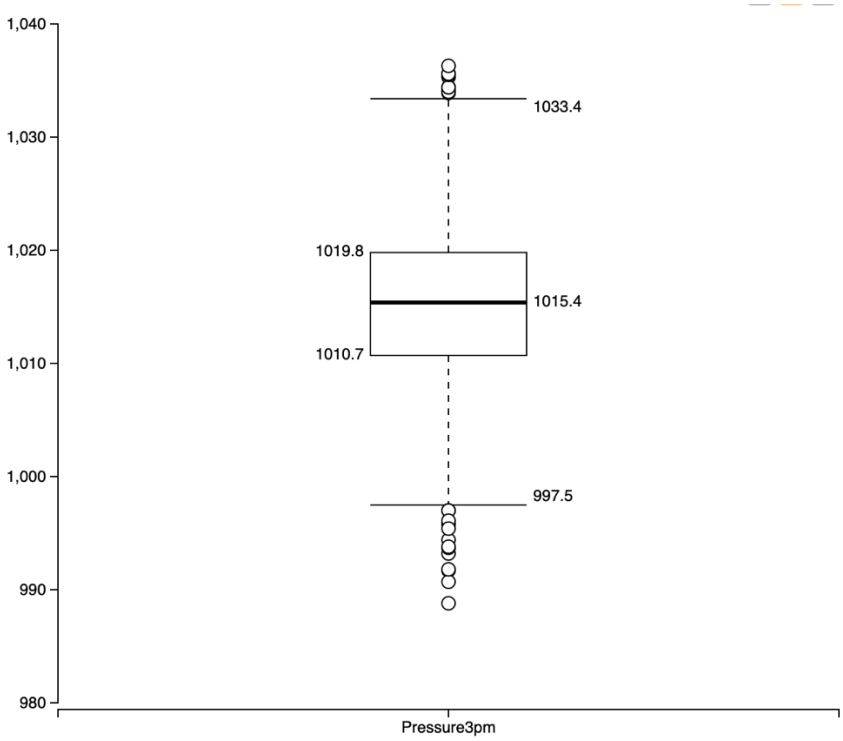


Figure 17: Box plot for Pressure3pm

The box plot of Pressure3pm indicates a median value of 1015.4 hPa. The interquartile range (IQR) extends from 1010.7 hPa to 1019.8 hPa, showing that the majority of the pressure data at 3 pm lies within this range. The maximum observed value reaches 1033.4 hPa, with some outliers below 997.5 hPa, indicating similar variability in atmospheric pressure in the afternoon, though slightly lower than the morning readings.

Statistics	Value
Mean	1015.415 hPa
Median	1015.4 hPa
Minimum Value	988.8 hPa
Maximum Value	1036.3 hPa
25% Quantile	1010.7 hPa
75% Quantile	1019.8 hPa
Standard Deviation	6.954 hPa
Variance	48.363

2.18 Cloud9am (Ratio)

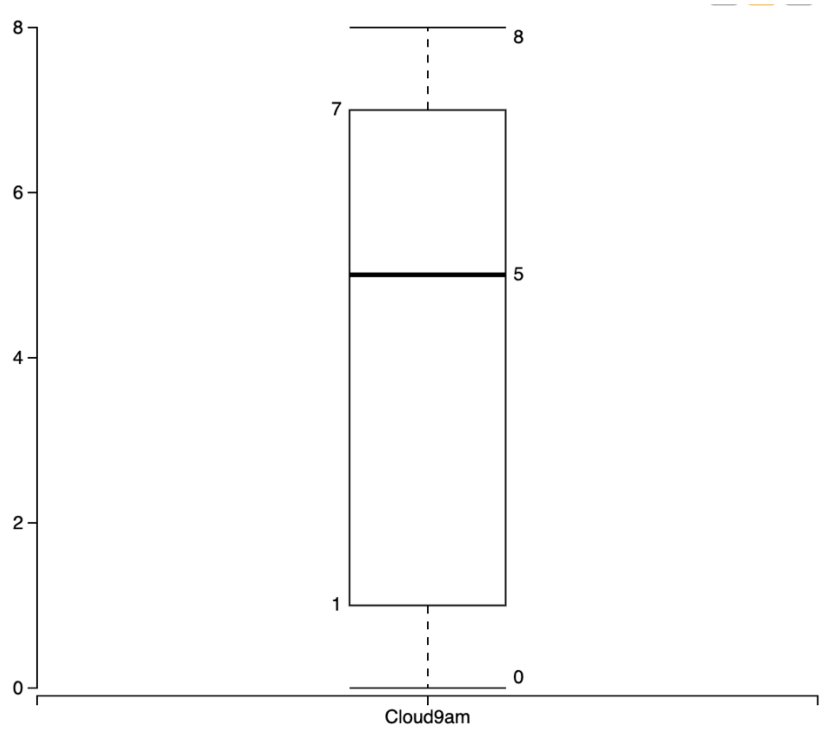


Figure18: Box plot for Cloud9am

The Cloud9am box plot reveals a median cloud cover of 5 oktas, similar to the 3 PM data. The interquartile range is from 1 okta to 7 oktas, indicating the spread of morning cloud cover data. Like the Cloud3pm plot, there are no extreme outliers, with data ranging from clear skies (0 oktas) to full cloud cover (8 oktas), demonstrating a stable cloud cover pattern in the mornings

Statistics	Value
Mean	4.521
Median	5
Minimum Value	0
Maximum Value	8
25% Quantile	1
75% Quantile	7
Standard Deviation	2.91
Variance	8.467

2.19 Cloud3pm (Ratio)

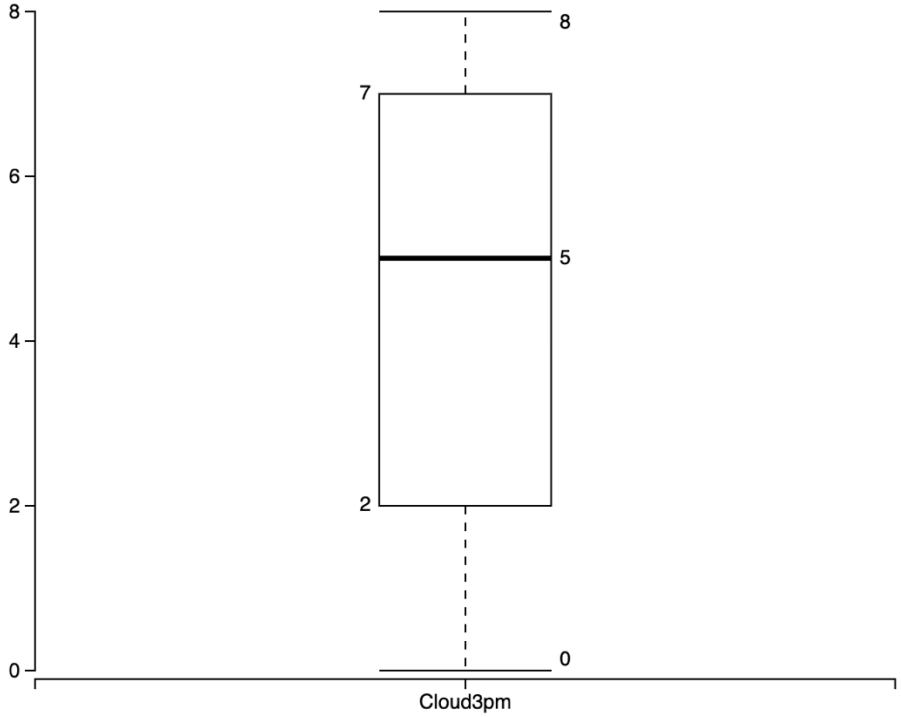


Figure 19: Box plot for Cloud3pm

The box plot for Cloud3pm indicates a median cloud cover of 5 oktas, where 1 okta represents one-eighth of the sky covered by clouds. The interquartile range extends from 2 oktas to 7 oktas, capturing the central tendency of cloud cover in the afternoon. The range is relatively wide, with some data points showing no clouds at all (0 oktas) and others at full cloud cover (8 oktas). There are no extreme outliers, suggesting a consistent pattern of cloud cover.

Statistics	Value
Mean	4.447
Median	5
Minimum Value	0
Maximum Value	8
25% Quantile	2
75% Quantile	7
Standard Deviation	2.748
Variance	7.551

2.20 Temp9am (Interval)

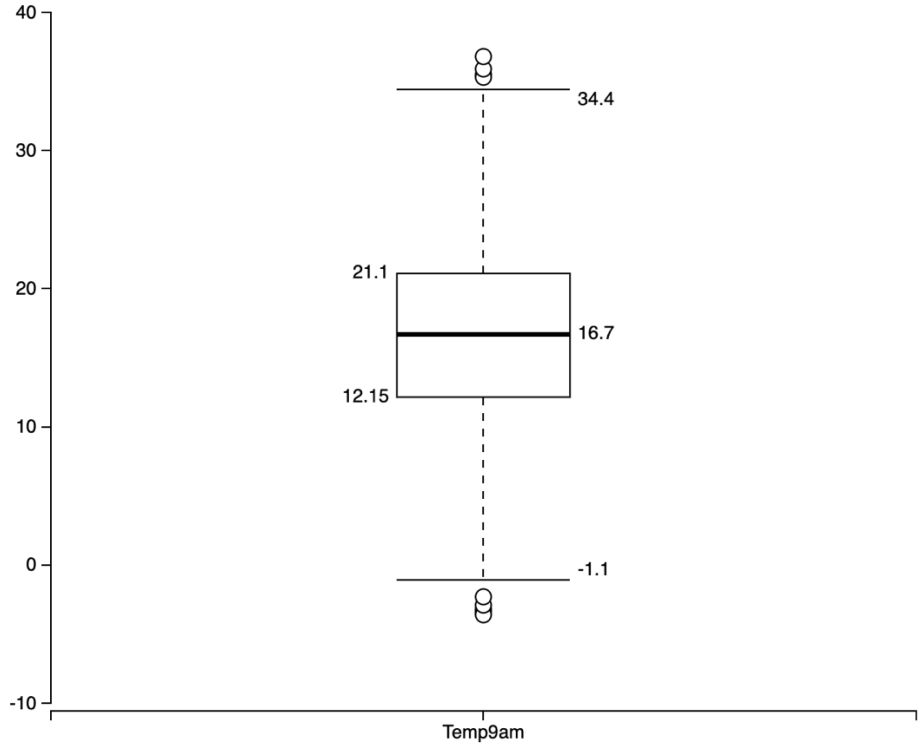


Figure 20: Box plot for Temp9am

The Temp9am box plot shows a median temperature of 16.7°C. The interquartile range is between 12.15°C and 21.1°C, which encompasses the majority of the temperature observations at 9 AM. Notably, there are outliers on both the higher side (above 34.4°C) and the lower side (as low as -1.1°C), indicating days with unusually high or low morning temperatures.

Statistics	Value
Mean	16.687
Median	16.7
Minimum Value	-3.6
Maximum Value	36.8
25% Quantile	12.125
75% Quantile	21.1
Standard Deviation	6.478
Variance	41.962

2.21 Temp3pm (Interval)

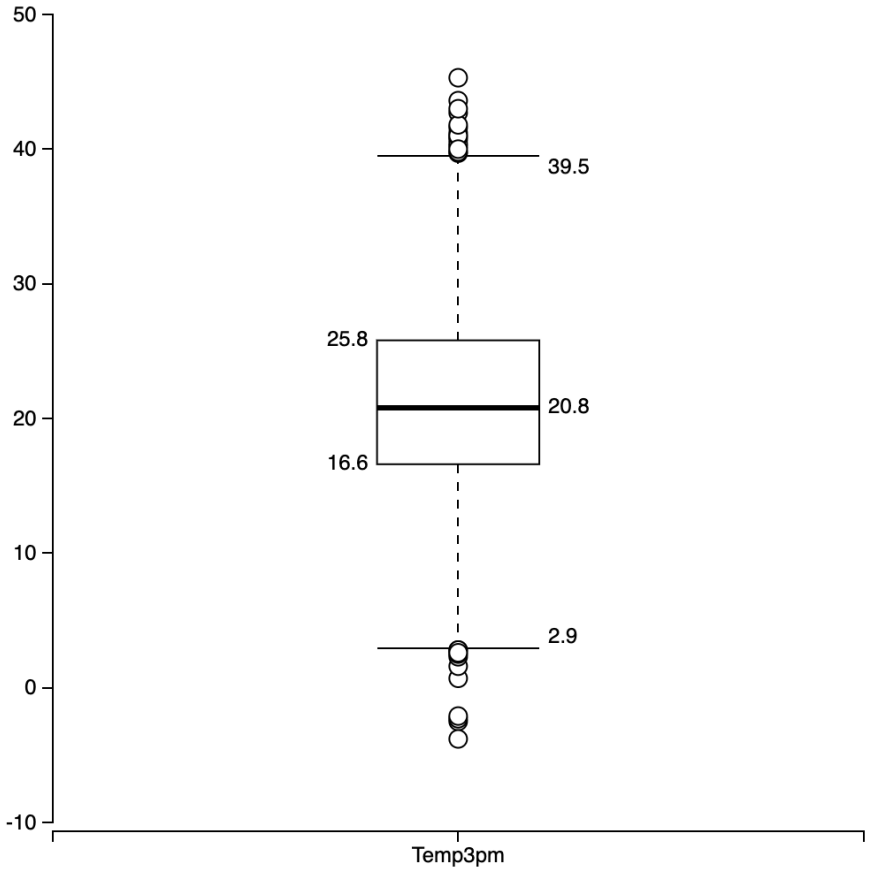


Figure 21: Box plot for Temp3pm

The box plot of Temp3pm indicates a median temperature of 20.8°C. The interquartile range (IQR) spans from 16.6°C to 25.8°C, reflecting the central spread of the data. The plot also shows a few extreme outliers, particularly above 39.5°C, indicating some days with significantly higher temperatures. The lower outliers around 2.9°C suggest a few exceptionally cooler days in the dataset.

Statistics	Value
Mean	21.364
Median	20.8
Minimum Value	-3.8
Maximum Value	45.3
25% Quantile	16.6
75% Quantile	25.8
Standard Deviation	6.841
Variance	48.806

2.22 RainToday (Nominal)

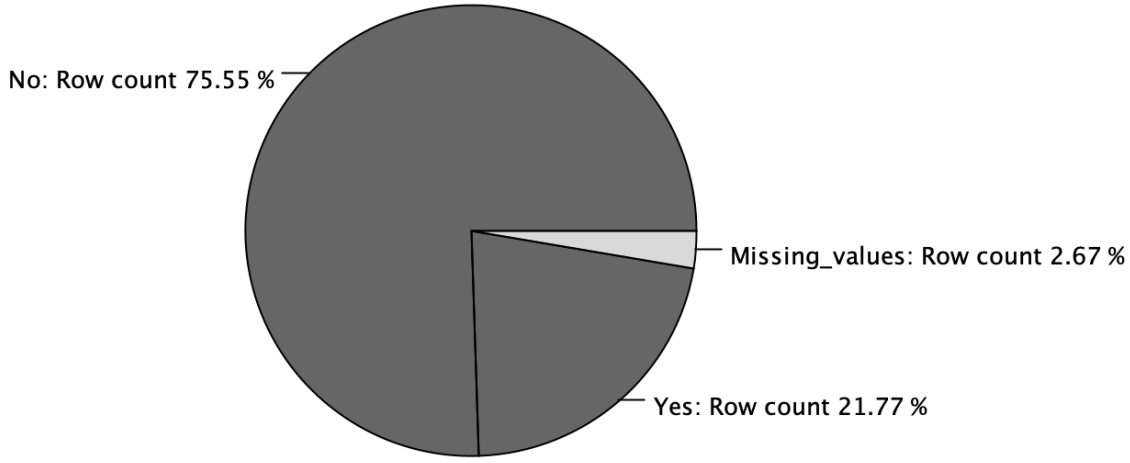


Figure 22: Pie chart for RainToday

The pie chart for RainToday illustrates the distribution of whether it rained today across the dataset. The majority of the observations, 75.55%, indicate that it did not rain today (No). A smaller portion, 21.77%, shows that it did rain today (Yes). Additionally, there is a 2.67% of missing data, representing entries where this information was not recorded. This distribution suggests that on most days in the dataset, there was no rainfall, but a significant portion of days did experience rain, with some data points lacking information.

2.23 RainTomorrow (Nominal)

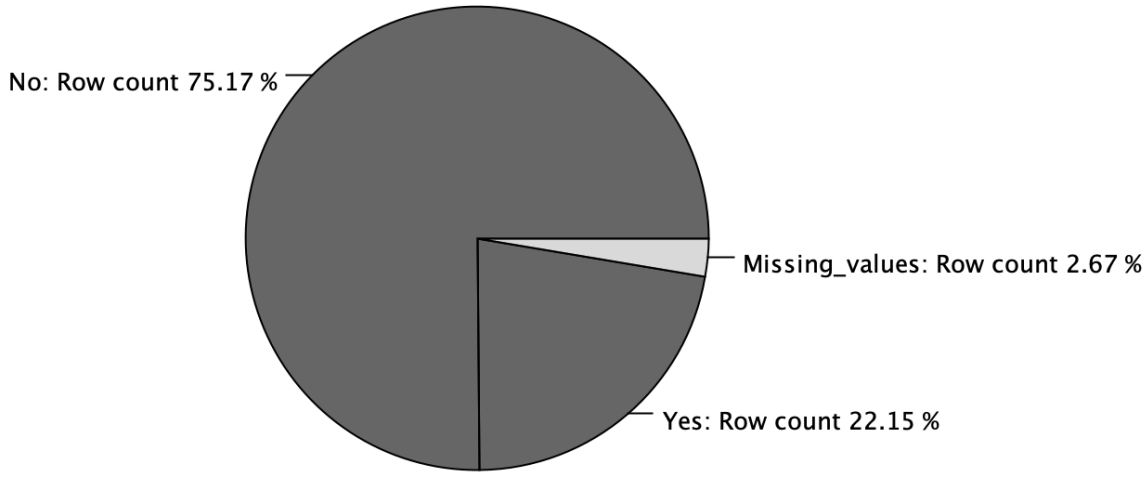


Figure 23: Pie chart for RainTomorrow

The pie chart for RainTomorrow displays the likelihood of rain occurring the next day based on the dataset. The largest portion, 75.17%, represents predictions where no rain (No) is expected. A smaller segment, 22.15%, indicates that rain (Yes) is anticipated. There is also a 2.67% portion of missing data, which accounts for the cases where the prediction is unavailable. This distribution suggests that the forecast predicts no rain for the majority of days, but there is still a notable percentage predicting rain, with a small fraction of the data missing.

3. Exploration

Finding



Figure 24: Scatter plot for relationship between Rainfall and RainTomorrow

This scatter plot reveals a key finding: rainfall above 50mm is often linked with a "Yes" prediction for rain the following day (RainTomorrow). Interestingly, even when no rain is predicted, the data shows instances of substantial rainfall, highlighting potential gaps in forecasting accuracy. Notably, the plot identifies outliers—days with unexpected heavy rainfall despite a "No" prediction. These outliers could indicate extreme weather events or forecasting errors. This finding underscores the challenge of predicting weather patterns accurately and suggests that current models may benefit from further refinement to better capture these anomalies

- **Outlier**

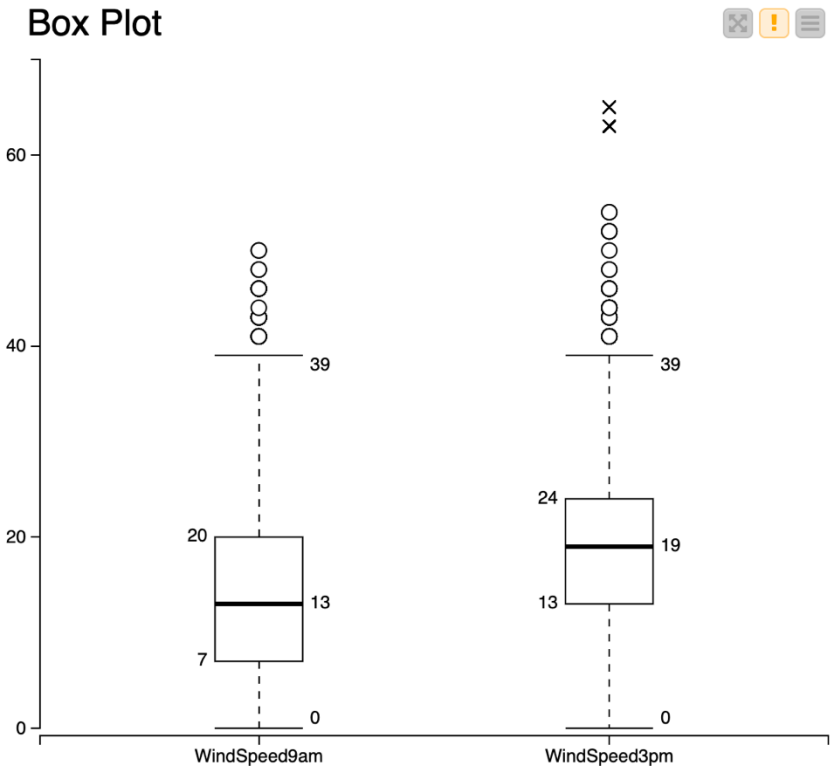


Figure 25: Box plot for WindSpeed9am and WindSpeed3pm

This box plot of WindSpeed9am and WindSpeed3pm reveals interesting outliers. Both distributions show a median wind speed of 13 km/h at 9am and 19 km/h at 3pm, respectively. However, there are significant outliers present in both datasets, particularly at the upper end, where wind speeds exceed 39 km/h. These outliers suggest occasional instances of unusually high wind speeds, which could be due to extreme weather conditions such as storms. The consistency in outliers across both times indicates that these high wind events are not isolated but occur throughout the day. This finding is important for weather modeling and forecasting, as it highlights the potential for extreme wind events that may not be well-represented by average conditions

Correlation

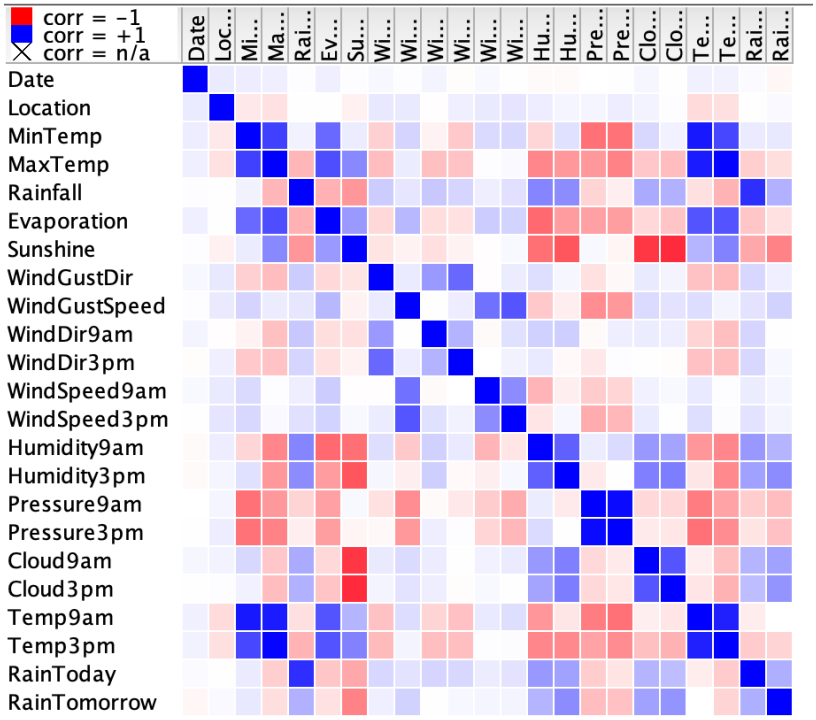


Figure 26: Rank Correlation graph

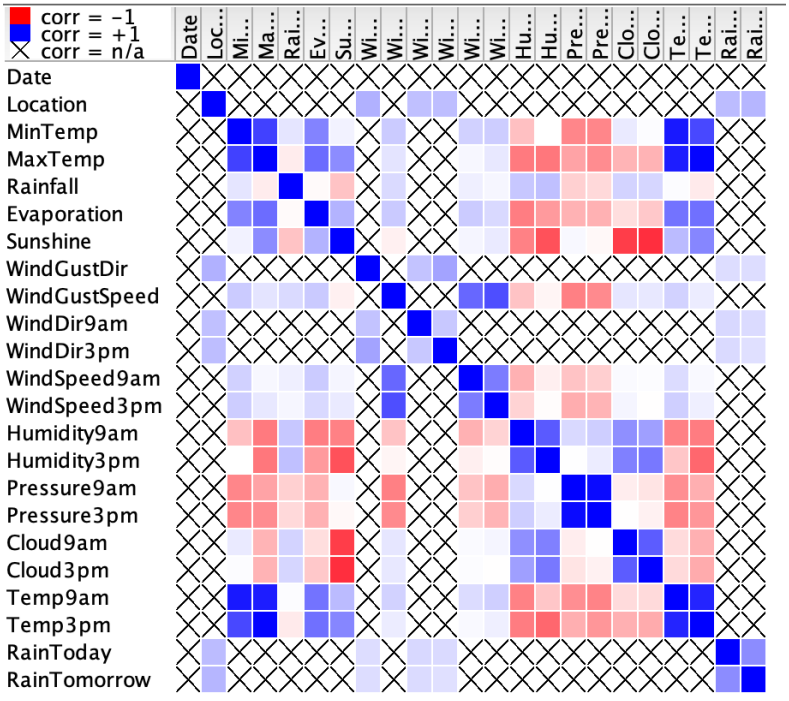


Figure 27: Linear Correlation graph

The Rank correlation matrix (Figure 26) and Linear correlation matrix (Figure 27) provides an overview of the relationships between various weather attributes in the dataset. The color shows the strength and direction of these correlations, blue signifies positive correlations, while red indicates negative correlations. These matrices reveal several interesting findings among the attributes, which could be highly beneficial for extracting meaningful insights.

1. **WindSpeed and WindGustSpeed**: There is a strong positive correlation between WindSpeed (both 9am and 3pm) and WindGustSpeed. This suggests that higher wind speeds tend to coincide with stronger wind gusts, which is expected in weather patterns.
2. **Temperature and Sunshine**: MaxTemp and Sunshine show a positive correlation, indicating that higher temperatures are often associated with sunnier days. This is logical as more sunshine typically leads to increased temperatures.
3. **Humidity and Temperature**: There is a noticeable negative correlation between Humidity (9am and 3pm) and MaxTemp/MinTemp. This suggests that higher temperatures are generally associated with lower humidity levels, which is common in many climates as warmer air can hold more moisture, reducing relative humidity.
4. **Cloud Cover and Sunshine**: As expected, Cloud (9am and 3pm) shows a negative correlation with Sunshine, indicating that days with more cloud cover tend to have less sunshine.
5. **Pressure and Temperature**: Pressure at 9am and 3pm shows a moderate negative correlation with Temperature. This suggests that higher temperatures are often associated with lower pressure systems, which could indicate the presence of warmer, more unstable air masses.
6. **Rainfall and RainTomorrow**: Rainfall shows a positive correlation with RainTomorrow, indicating that higher rainfall today increases the likelihood of rain tomorrow, which aligns with typical weather patterns

Clustering

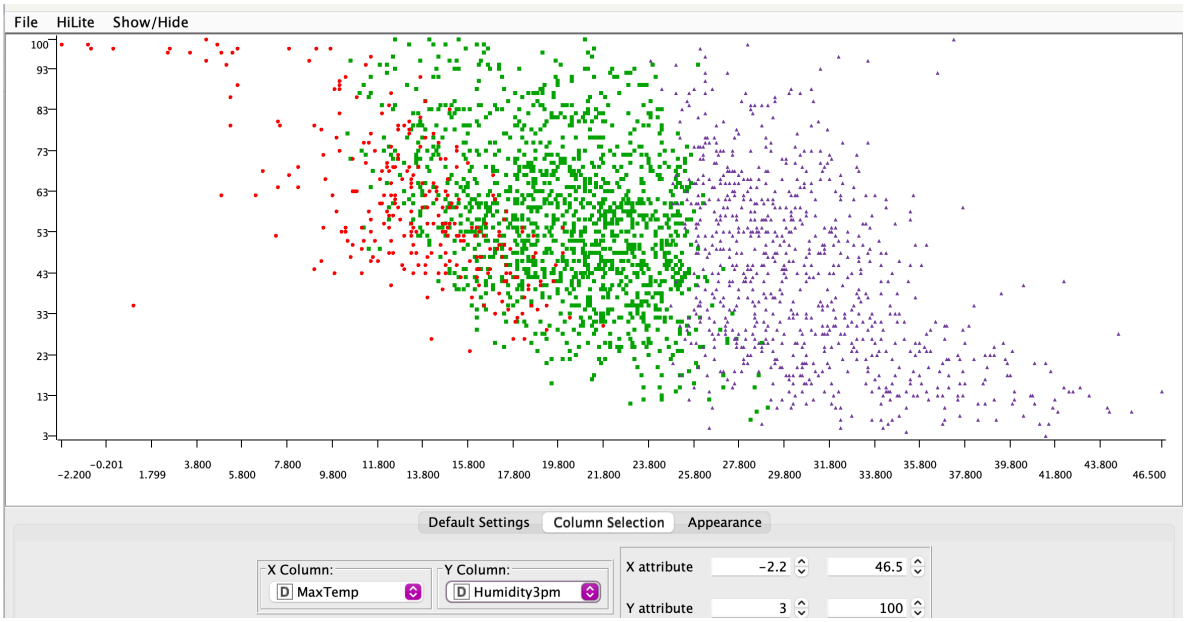


Figure 28: Clustering Analysis of MaxTemp and Humidity3pm

In this analysis (Figure 28), it has been conducted clustering on the Maximum Temperature (MaxTemp) and Humidity at 3 PM (Humidity3pm) attributes to uncover patterns and relationships between these variables. The data was successfully segmented into three distinct clusters, each characterized by unique temperature and humidity profiles.

Finding

Correlation Between Humidity and Temperature: The clustering reveals a distinct inverse correlation between temperature and humidity, where higher temperatures are generally associated with lower humidity levels, and vice versa. This is a typical meteorological pattern, as warmer air can hold more moisture, but when temperatures rise significantly, the relative humidity tends to decrease if additional moisture is not added to the air.

B. Data Pre-preprocessing

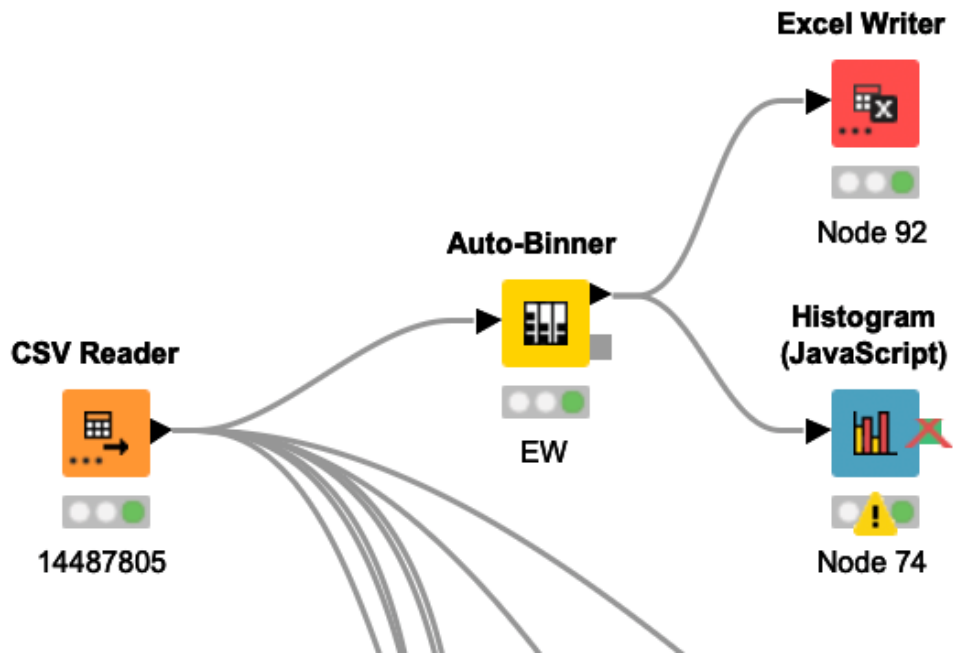
1. Binning techniques of Rainfall Attribute

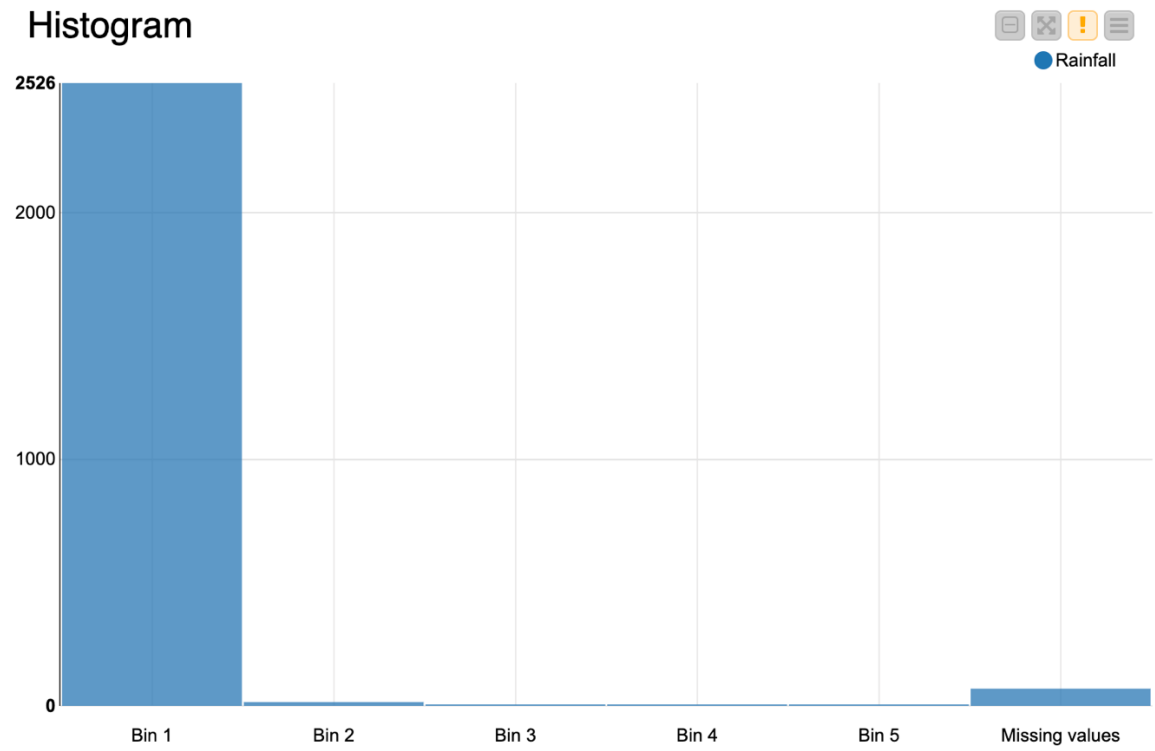
Binning is used to smooth noisy data and identify patterns by grouping continuous values into discrete intervals or "bins." This process helps in reducing the complexity of the data and in visualizing distributions more effectively.

1.1 Equi-width binning

The equi-width binning technique divides the Rainfall attribute into intervals of equal width. The resulting histogram shows a significant concentration of data in the first bin, which indicates that most rainfall values are low, with very few instances of higher rainfall amounts.

- **Knime process for Equi-width**





The equi-width binning technique divides the Rainfall attribute into intervals of equal width. The resulting histogram shows a significant concentration of data in the first bin, which indicates that most rainfall values are low, with very few instances of higher rainfall amounts.

- **Binned data for Equi-width**

Row ID	Rainfa...	Rainfall	
Row0	Bin 1	0	2
Row1	Bin 1	0	2
Row2	Bin 1	0	2
Row3	Bin 1	0	2
Row4	?	?	2
Row5	Bin 1	3.4	2
Row6	Bin 1	0	2
Row7	Bin 1	0	2
Row8	Bin 1	0.8	2
Row9	Bin 1	0	2
Row10	Bin 1	0	2
Row11	Bin 1	0	2
Row12	Bin 1	0.2	2
Row13	Bin 1	0	2
Row14	Bin 1	0	2
Row15	Bin 1	0	2
Row16	Bin 1	4	2
Row17	Bin 1	0	2
Row18	?	?	2
Row19	Bin 1	1	2
Row20	Bin 1	0	2
Row21	Bin 1	0	2

☐ Excel for Equi-width

	A	B	C
1	Rainfall	EW	ED
2	0	Bin 1	Bin 1
3	0	Bin 1	Bin 1
4	0	Bin 1	Bin 1
5	0	Bin 1	Bin 1
6			
7	3.4	Bin 1	Bin 5
8	0	Bin 1	Bin 1
9	0	Bin 1	Bin 1
10	0.8	Bin 1	Bin 2
11	0	Bin 1	Bin 1
12	0	Bin 1	Bin 1
13	0	Bin 1	Bin 1
14	0.2	Bin 1	Bin 2
15	0	Bin 1	Bin 1

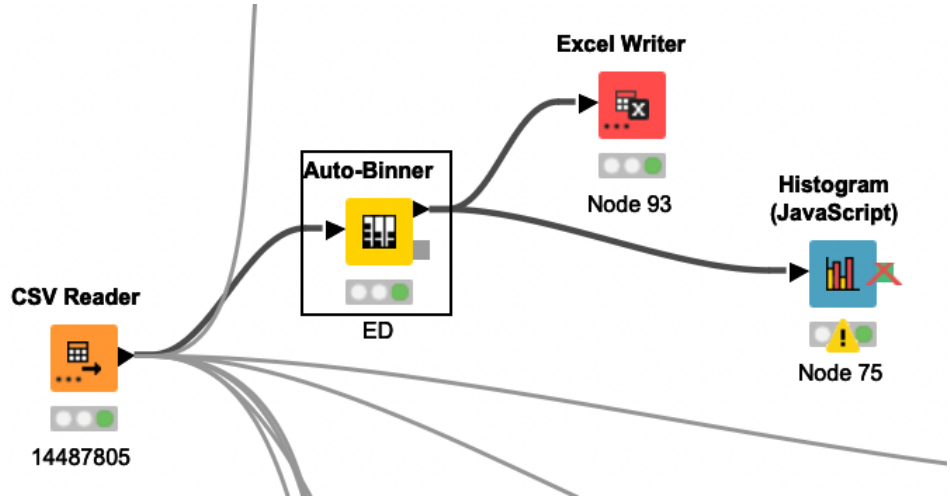
☐ Interpretation

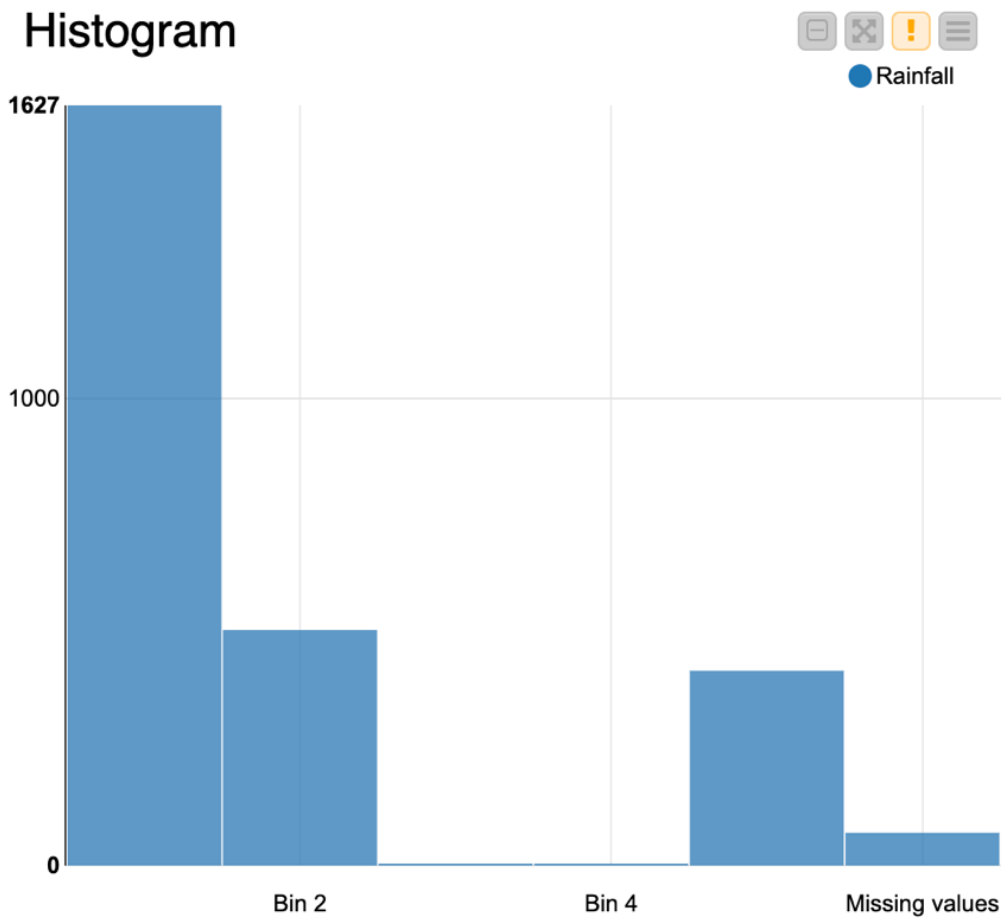
This pattern suggests that the dataset predominantly contains days with low rainfall, highlighting the rarity of heavy rain events.

1.2 Equi-depth binning

The equi-depth binning technique creates bins where each bin contains an equal number of data points from the Rainfall attribute. The histogram for this technique displays a more balanced distribution across the bins, indicating the presence of distinct clusters within the rainfall data.

☐ Knime process for Equi-depth





The histogram for equi-depthbinning of Rainfall attribute displays a more balanced distribution across the bins, indicating the presence of distinct clusters within the rainfall data.

- **Binned data for Equi-depth**

Row ID	S Rainfa...	D Rainfall
Row1	Bin 1	0
Row2	Bin 1	0
Row3	Bin 1	0
Row4	?	?
Row5	Bin 5	3.4
Row6	Bin 1	0
Row7	Bin 1	0
Row8	Bin 2	0.8
Row9	Bin 1	0
Row10	Bin 1	0
Row11	Bin 1	0
Row12	Bin 2	0.2
Row13	Bin 1	0
Row14	Bin 1	0
Row15	Bin 1	0

- **Excel for Equi-depth**

	A	B	C
1	Rainfall	EW	ED
2	0	Bin 1	Bin 1
3	0	Bin 1	Bin 1
4	0	Bin 1	Bin 1
5	0	Bin 1	Bin 1
6			
7	3.4	Bin 1	Bin 5
8	0	Bin 1	Bin 1
9	0	Bin 1	Bin 1
10	0.8	Bin 1	Bin 2
11	0	Bin 1	Bin 1
12	0	Bin 1	Bin 1
13	0	Bin 1	Bin 1
14	0.2	Bin 1	Bin 2
15	0	Bin 1	Bin 1

- **Interpretation**
  This technique provides a more detailed view of how rainfall is distributed, indicating that there are specific ranges where rainfall is more common, allowing for better insights into the variability of rainfall within the dataset.

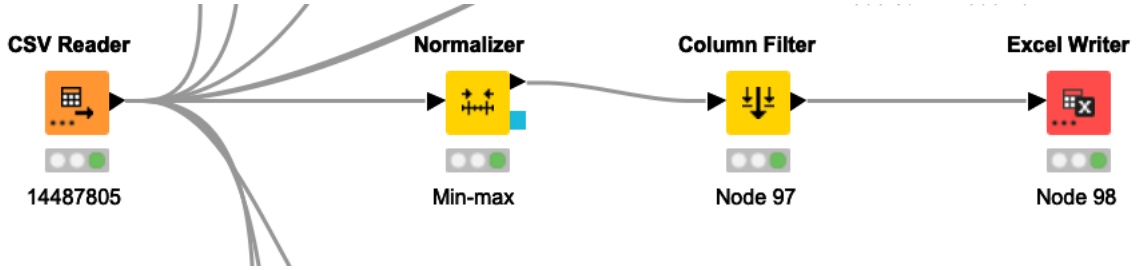
2. Normalisation of Maxtemp Attribute

Normalization is a crucial preprocessing step in data analysis that involves adjusting the values of numerical attributes to a common scale without distorting the differences in the ranges of the values. This process is particularly important when the data includes attributes like MaxTemp.

2.1 Min-Max

Min-Max Normalization scales the MaxTemp attribute to a fixed range, [0, 1]. This method is useful when the distribution of the MaxTemp data does not follow a normal distribution and makes it easier to compare with other attributes.

- **Knime process for Min-Max Normlisation**



- **Normalised Table of MaxTemp normalisation with Min-Max**

Row ID	MaxTemp
Row0	0.63
Row1	0.556
Row2	0.509
Row3	0.616
Row4	?
Row5	0.515
Row6	0.505
Row7	0.604
Row8	0.361
Row9	0.505
Row10	0.719

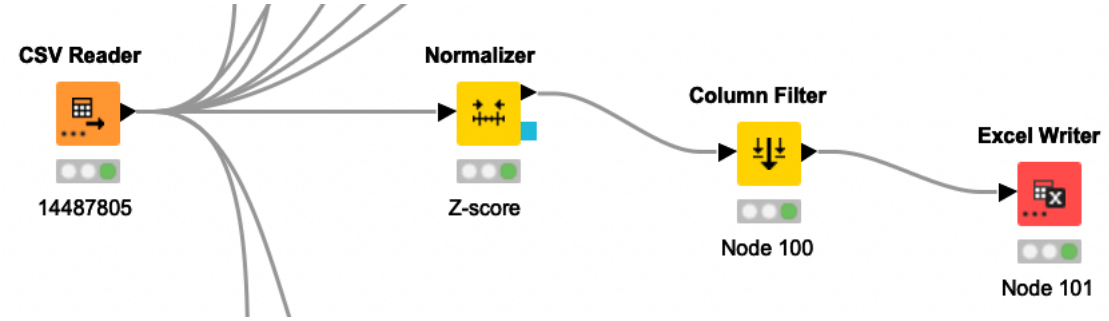
- **Interpretation**

The table illustrates the normalized MaxTemp attribute, where all values are scaled between 0 and 1. This transformation ensures that no value in the dataset dominates due to its larger scale, allowing for a balanced analysis across attributes.

2.2 Z-score

Z-score Normalization (or Standardization) transforms the MaxTemp attribute to have a mean of 0 and a standard deviation of 1 by subtracting the mean of the attribute and dividing by the standard deviation

- **Knime process for Z-score normalisation**



- **Normalised Table of MaxTemp normalisation with Z-score**

Row ID	MaxTemp
Row0	0.797
Row1	0.287
Row2	-0.038
Row3	0.698
Row4	?
Row5	0.004
Row6	-0.067
Row7	0.613
Row8	-1.058
Row9	-0.067
Row10	1.406

- **Interpretation**

The table shows the Z-score normalized values of MaxTemp, where the data points are now represented by how many standard deviations they are away from the mean.

3. Discretisation of WindSpeed3m Attribute

Discretization is a data preprocessing technique that involves transforming continuous data into discrete intervals or categories. In this case, for the WindSpeed3pm attribute, the discretization process categorized the wind speeds into four below distinct categories

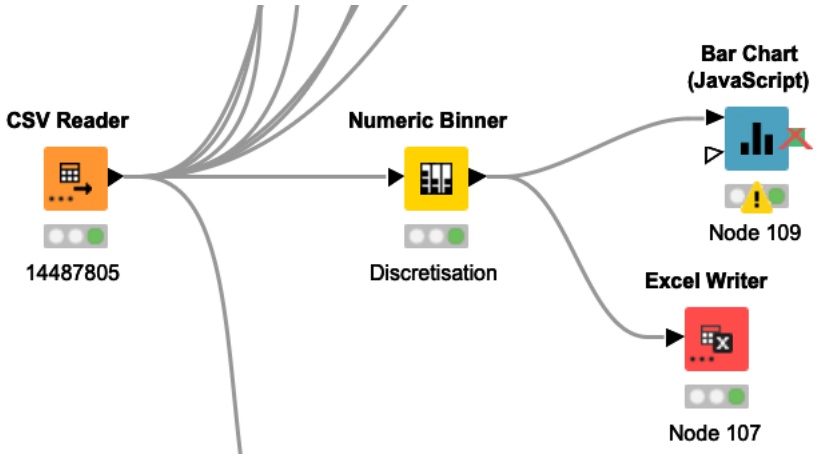
1-Slow Wind: Wind speeds less than or equal to 13.0 units.

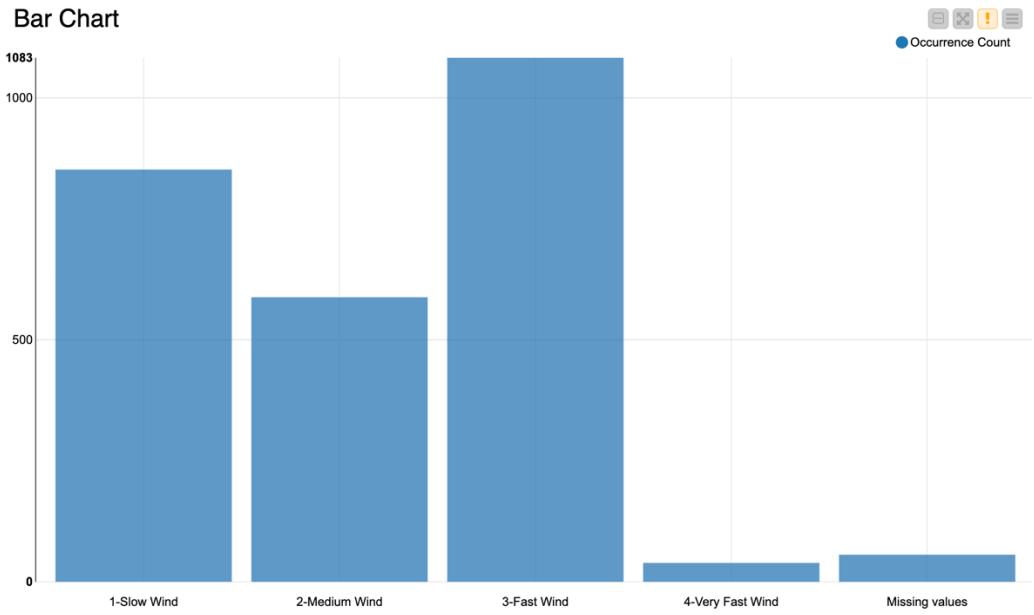
2-Medium Wind: Wind speeds greater than 13.0 and up to 19.0 units.

3-Fast Wind: Wind speeds greater than 19.0 and up to 39.0 units.

4-Very Fast Wind: Wind speeds greater than 39.0 units.

- **Knime process for Discretisation of WindSpeed3pm**





- Fast Wind: The most frequent category, indicating that wind speeds categorized as "Fast" occurred more often in the dataset.

- Slow Wind: The second most common category, showing that "Slow" wind speeds are also prevalent.

- Medium Wind: This category appears less frequently than "Slow" and "Fast" winds.

- Very Fast Wind: This is the least common category, suggesting that very high wind speeds are rare in the dataset.

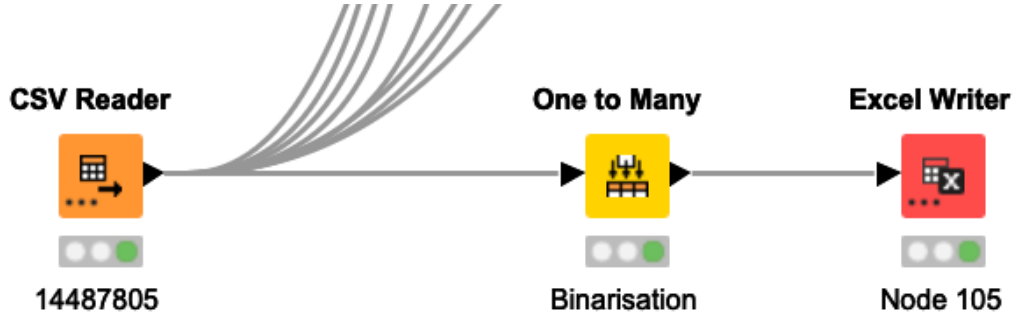
- **Discretisation data for WindSpeed3pm**

Row ID	WindSpeed3pm
Row0	1-Slow Wind
Row1	3-Fast Wind
Row2	2-Medium Wind
Row3	1-Slow Wind
Row4	?
Row5	4-Very Fast Wind
Row6	3-Fast Wind
Row7	3-Fast Wind
Row8	3-Fast Wind
Row9	2-Medium Wind
Row10	2-Medium Wind

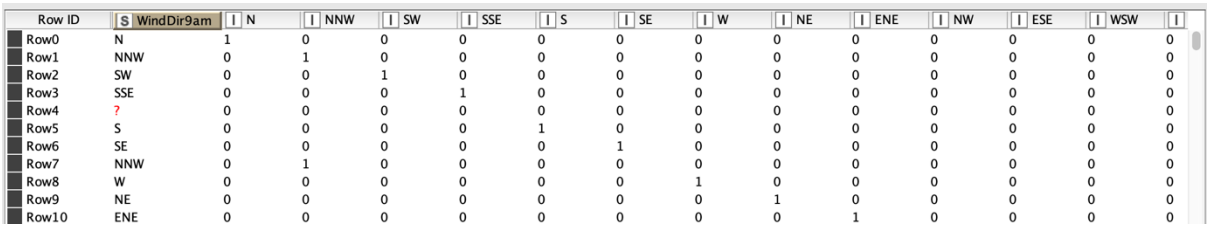
4. Binarisation of WindDir9am Attribute

Binarisation is the process of converting categorical data into a binary form, which is useful when preparing data for machine learning models that require numerical input

- **Knime process for Binarisation of WindDir9am Attribute**



Binarisation data of WindDir9am Attribute



Row ID	WindDir9am	N	NNW	SW	SSE	S	SE	W	NE	ENE	NW	ESE	WSW	
Row0	N	1	0	0	0	0	0	0	0	0	0	0	0	0
Row1	NNW	0	1	0	0	0	0	0	0	0	0	0	0	0
Row2	SW	0	0	1	0	0	0	0	0	0	0	0	0	0
Row3	SSE	0	0	0	1	0	0	0	0	0	0	0	0	0
Row4	?	0	0	0	0	0	0	0	0	0	0	0	0	0
Row5	S	0	0	0	0	1	0	0	0	0	0	0	0	0
Row6	SE	0	0	0	0	0	1	0	0	0	0	0	0	0
Row7	NNW	0	1	0	0	0	0	0	0	0	0	0	0	0
Row8	W	0	0	0	0	0	0	1	0	0	0	0	0	0
Row9	NE	0	0	0	0	0	0	0	1	0	0	0	0	0
Row10	ENE	0	0	0	0	0	0	0	0	1	0	0	0	0

In this case, the *WindDir9am* attribute, which originally contained categorical wind direction values (like N, SSE, etc.), was transformed into multiple binary columns. Each column represents a specific wind direction, with "1" indicating the presence of that direction and "0" indicating its absence.

C. Summary

1. Key Findings

1. **Rainfall Distribution and Predictions**
    - The binning analysis of the *Rainfall* attribute revealed that most data points fall within the lower rainfall ranges. However, equi-depth binning identified distinct clusters in the rainfall data, suggesting variability in rainfall patterns that could be further examined to understand seasonal or regional differences.
2. **WindSpeed Analysis**
    - The *WindSpeed3pm* attribute was discretized into four categories: *Slow Wind*, *Medium Wind*, *Fast Wind*, and *Very Fast Wind*. The data predominantly falls into the *Fast Wind* category, with fewer occurrences in the *Very Fast Wind* category, indicating that extremely high wind speeds are relatively uncommon.
3. **Directional Trends in Wind**
    - The binarization of the *WindDir9am* attribute revealed a distribution of wind directions, with certain directions such as NNW and E appearing more frequently. This trend could have implications for regional weather patterns and should be considered in further analyses.
4. **Temperature Normalization**
    - The *MaxTemp* attribute was normalized using both Min-Max and Z-score techniques. The Min-Max normalization highlighted the range of temperatures within a 0-1 scale, while the Z-score provided a standardized view, showing

how temperatures deviate from the mean, which is useful for comparative analyses across different datasets.

2. Additional Information & Business Insights

- Potential Clusters and Associations
  - The clustering observed in the rainfall data and the distribution of wind speed categories suggest underlying patterns or associations that warrant further investigation. For example, correlations between high rainfall and specific wind directions could be explored to enhance weather prediction models.

3. Possible Business Questions within the given dataset

Q1. How can the observed wind speed distributions be leveraged to optimize wind energy production in specific regions?

Q2. What factors contribute to the clustering in rainfall data, and how can this be utilized to improve agricultural planning?

Q3. How can the binarized wind direction data be used to enhance logistics planning for industries that are weather-dependent?

4. Interesting Findings

- The majority of rainfall data is concentrated in lower ranges, but distinct clusters identified through equi-depth binning suggest potential seasonal variations or localized weather events.
- The *Fast Wind* category is predominant in the *WindSpeed3pm* attribute, indicating a trend that could have significant implications for sectors like aviation or wind energy